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The 2025 Long-term Fiscal Model: Main features and changes

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Executive summary

The *Public Finance Act 1989* requires the Treasury to prepare a Long-term Fiscal Statement (LTFS) on the potential long-term fiscal outlook for New Zealand. There have been six Statements to date (2006, 2009, 2013, 2016, 2021 and 2025), with a requirement for each to be produced no more than four years since the previous edition[1] and to cover a minimum of the next 40 years. Every LTFS has utilised a version of the Long-term Fiscal Model (LTFM) to produce some of the projections described and/or illustrated in it. The main purpose of the LTFM is to project New Zealand’s long-term fiscal position, under a given set of assumptions, so that potential future fiscal pressures can be identified and better understood.

The 2021 LTFS included with its publication a background paper titled *Demographic, Economic and Fiscal Assumptions and Logic in the 2021 Long-term Fiscal Model* (New Zealand Treasury, 2021). That paper provided a detailed description of the 2021 version of the LTFM. Many of the projection techniques for various demographic, economic and fiscal variables are unchanged from the 2021 model in the 2025 LTFM, which has as its base the forecasts produced for the 2025 Budget Economic and Fiscal Update (New Zealand Treasury, 2025a). Rather than repeat those descriptions in detail, this paper focuses on important aspects of the projections that have changed since the 2021 version was produced. Readers who would like more detail about any areas of the projections are encouraged to find this information in the 2021 background paper.

[1] The 2021 LTFS was published five years after its predecessor, due to difficulties caused by the COVID-19 pandemic. Parliament approved this postponement, acknowledging it to be the result of an unusual set of circumstances.

The LTFM's base projection models *net core Crown debt*, as a percentage of *nominal gross domestic product* (GDP), rising at an accelerating rate. This illustrates the main areas of fiscal pressure caused by an ageing population structure, climate change, and other factors, over coming decades. Once these pressures cause expenditure to overtake revenue, the resulting deficits are funded through increased borrowing. As debt grows it generates increased interest expenses, which further add to both deficits and borrowings. Such a base projection has been a feature of every LTFS produced to date, although when the lift in debt begins, and the level of GDP reached after 40 years, has varied with the forecast base and the projection assumptions.

The modelled economic and fiscal variables of the LTFM, whether they are components of the baseline projection or of alternative scenarios, should not be viewed as forecasts. The latter term refers to attempts to predict future outcomes by using wide-ranging resources, comprehensive modelling and expert opinion and knowledge. Projections, on the other hand, illustrate potential outcomes that result under a given set of assumptions, where these are often based on historical averages or existing policy settings.

A particular feature of the LTFM projections is that they are based upon no policy changes introduced to address fiscal pressures, but rather successive governments just borrowing more to meet them. This is unrealistic for several reasons.

Among the most important are:

- New Zealand's public finance settings require governments to set prudent debt objectives
- for the last 40 years New Zealand's public debt to GDP has mainly followed a decreasing pathway, other than when circumstances required increased borrowing for periods of a few years following the global financial crisis (GFC) and the COVID-19 pandemic, and
- overseas lenders would stop providing funds to any country that just allowed public debt to keep rising, with no obvious plans produced to show how the fiscal position will be corrected.

The LTFM's baseline scenario is most certainly not a forecast, but rather it is a warning signal of the need for policy changes to address the fiscal pressures caused by an ageing population, higher demand for health services, replacing infrastructure, and other factors.

There are many options available to avoid these undesirable outcomes. One of the key objectives of the Long-term Fiscal Statements is to stimulate discussion and debate concerning these options. The Long-term Fiscal Model serves as just one of many components that goes into producing each Long-term Fiscal Statement and assists in achieving these objectives.

Introduction

Key Points

- This paper mainly focuses on what has changed since the last version of the LTFM was produced for the 2021 LTFS.
- Readers who wish to know more details about all aspects of the LTFM are encouraged to read the 2021 background paper cited in this section.
- The LTFM's "exploding debt" scenario illustrates a warning signal, not a forecast, and is similar in logic to the long-term fiscal projections of several international agencies.

The 2021 background paper *Demographic, Economic and Fiscal Assumptions and Logic in the 2021 Long-term Fiscal Model* had an introduction section which covered the requirement for the Long-term Fiscal Statement, the role of the Long-term Fiscal Model, and some key points about the Long-term Fiscal Model.

Nothing in regard to the requirements for the LTFS have changed with this 2025 version, which is the sixth in the series. Ongoing fiscal pressures remain, not just because of New Zealand's ageing population structure, but also due to other long-term issues such as climate change. The need to draw attention to these pressures, and to put forward potential means of addressing them and mitigating their negative impacts, is still an important obligation for the Treasury.

The Long-term Fiscal Model (LTFM) (New Zealand Treasury, 2025b) does change from update to update, and one of this paper's main purposes is to describe the differences in the 2025 LTFM relative to the 2021 version of the model. However, the main role of the LTFM, as described in the 2021 background paper, has not fundamentally changed and remains largely the same as it has been since the first LTFM was published with the 2006 LTFS.

In summary, the LTFM is a spreadsheet model that uses demographic, economic and fiscal data to project out potential scenarios for New Zealand's public fiscal position over the ensuing four decades. Projections are produced using mathematical formulae and assumptions specific to the variable being projected. The assumptions are derived from historical growth rates, relationships to other variables, existing or planned policy, or a mixture of these influences.

The LTFM's projection where debt as a percentage of GDP[2] accelerates upward is again the baseline scenario produced by the 2025 LTFM. This scenario involves fiscal pressures, caused by an ageing population structure, climate change, replacing infrastructure, changing overseas influences, and other factors. These fiscal pressures result in annual expenses exceeding revenue early into the projected years. Once this occurs increased borrowing is used to fund the growing deficits. As debt levels rise, they produce increased interest costs, which lead to even bigger deficits and consequently accelerate the growth of borrowing.

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[2] Unless specifically stated otherwise, references in this paper to GDP should be taken to mean the current price measure, nominal gross domestic product, rather than the constant price, volume-based measure real GDP.

This “exploding debt” scenario, while not a likely outcome as it supposes no policy changes to address the fiscal pressures, is also used by other nation’s treasuries and finance ministries. Examples include projections illustrated in the *Long-term Budget Projections* of the United States Congressional Budget Office (Congressional Budget Office, 2025); the *Fiscal Risks and Sustainability Report* of the United Kingdom Office for Budget Responsibility (Office for Budget Responsibility, 2025); and the *Intergenerational Reports* of the Australian Treasury (Australian Treasury, 2023).

Past versions of the LTFS have included LTFM scenarios that model alternatives to the “exploding debt” outlook, including some that keep net core Crown debt to GDP at manageable levels. The 2025 LTFS also includes some alternative projections produced by the 2025 LTFM.

The key points about the LTFM described in the 2021 background paper are all still relevant too. The 2025 LTFM projects over a 41-year horizon from the last known outturn, or historical, fiscal (June-end) year of 2023/24. Strictly speaking only 36 of these fiscal years, from 2029/30 to 2064/65, are projections as the first five fiscal years from 2024/25 to 2028/29 are the Budget Economic and Fiscal Update 2025 forecast years. While 2024/25 had become an outturn year by the time that the 2025 LTFS was published, it was still incomplete and hence a forecast year when the Budget 2025 was produced in May 2025.

The entity coverage described in the 2021 background paper is unchanged, as is the basic structure of the main modelling worksheet. The manner in which assumptions are used to alter the pathways of projections, and how they are entered into the LTFM, are no different in the 2025 LTFM than they were in the 2021 model.

The differences between forecasts and projections have been covered in the Executive Summary, but it is important to keep these in mind when viewing the projections of the 2025 LTFM. To reiterate, forecasts are genuinely intended to predict future outcomes, whether or not they actually do a particularly good job of achieving this. The goal of producing projections is to illustrate potential paths, which are highly dependent on the assumptions behind them. They may not even be designed to represent a likely outcome but rather serve to indicate a future situation that must be averted through policy and behavioural responses. The baseline scenario of the 2025 LTFM, the “exploding debt” scenario, is just such a warning signal projection.

Demographic variables and assumptions

Key Points

- New Zealand, like many countries, has an ageing population structure.
- Both ends of the life cycle contribute to this change, through falling fertility and rising longevity. The latter is something to be celebrated, but it does bring fiscal challenges.
- The 2025 LTFM uses Stats NZ's latest population and labour force projections.

Demographics are an important part of the LTFM. Changes in age and gender distributions over time guide the paths of projections of various welfare expenses, including the public pension *New Zealand Superannuation*, *Health* expenditure and *Education* spending. The growth of the labour force from year to year is a key input into GDP projections.

It is for this reason that the 2021 background paper devoted a detailed section to describing the role of demographics in the LTFM. This material is still relevant, and nothing has significantly altered in how the *national population projections* (Statistics New Zealand, 2025a) and *national labour force projections* (Statistics New Zealand, 2025b), produced by Statistics New Zealand (Stats NZ), are implemented in the LTFM.

Both the population and labour force projections have been updated since the 2020 base year versions used in the 2021 LTFM. The latest LTFM applies population and labour force projections that were produced in June 2025 and have 2024 bases. As well as this, the 2023 Census, which informs both projections, has occurred and been analysed and collated since the 2021 LTFS was produced.

The two inputs into demographic projections that have significantly changed between the Stats NZ's 2020 base population projections and the latest 2024 base version are those of net migration and fertility. Both of these also affect the updated labour force projections.

Net migration has altered from assuming a set number in almost all years to a steadily increasing amount annually. As more than half of migrants are in the age group from 25 to 44 years inclusive this rising number strengthens labour force projections, and hence GDP.

Assumptions concerning the *fertility rate* over time have lowered compared to those made in the 2020 base population projections, so that those in the 2024 base population projections are even further from the replacement rate of 2.1 by which a population can stabilise without any net migration. This has the opposite effect to the increased *net migration* track, as it weakens labour force and GDP projections. This is because reducing projected *fertility rates* leads to lower numbers of people being born in any given year and hence growing up to be potentially included in the labour force in later projected years.

Table 1 below summarises some of the key differences between the population and labour force projections used in the 2021 and 2025 versions of the LTFM. The starting point for most of the statistics listed in Table 1 for the 2021 LTFM is 2019/20, which was the base year of the demographic and labour force projections used at that time. For the 2025 LTFM the starting point in Table 1 is 2023/24, the base year of Stats NZ's latest projections. While the 2021 LTFM provided economic and fiscal projections only as far as 2060/61, its demographic and labour force inputs to 2072/73 were included in the model.

Table 1: Differences between 2021 and 2025 demographic and labour force projections

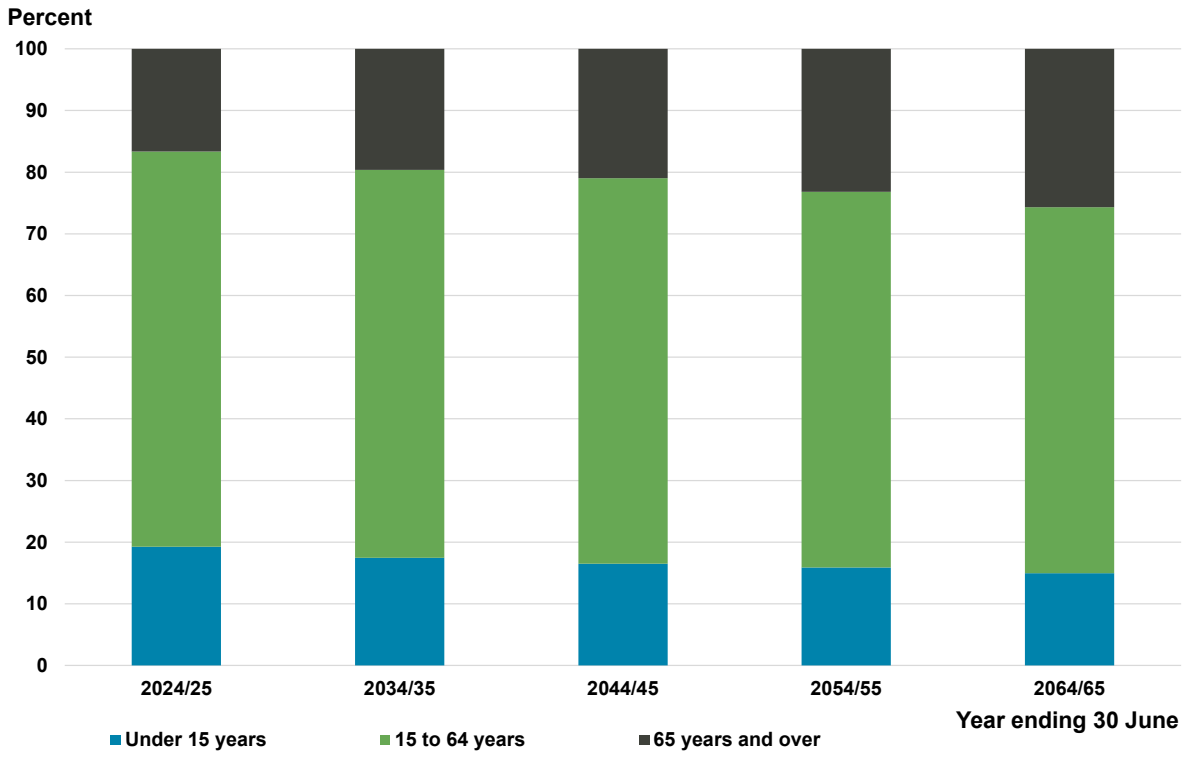
Demographic or labour force variable	2021 LTFM	2025 LTFM
Fertility rate (births per woman of childbearing age)	Stabilises at 1.65 throughout the projection	Reduces over initial 19 years from 1.59 to stabilise at 1.55
Life expectancy at birth	Rises from 80.8 to 86.6 years in 2064/65 for males & from 84.4 to 89.5 years in 2064/65 for females	Rises from 80.8 to 86.0 years in 2064/65 for males & from 84.4 to 89.2 years in 2064/65 for females
Life expectancy at age 65 years	Rises from 19.7 to 23.6 years in 2064/65 for males & from 22.2 to 25.8 years in 2064/65 for females	Rises from 19.9 to 23.8 years in 2064/65 for males & from 22.1 to 25.7 years in 2064/65 for females
Net migration per year	First two projected years assume 10,000 and 20,000 respectively and then stabilises at 25,000 for all remaining years of the projections	Rises from 25,000 to 36,000 between 2024/25 and 2028/29 and then by 1,000 every 4 years after that to reach 46,000 by 2064/65
Median age in years	Lifts from 37.4 to 46.3 in 2064/65	Lifts from 37.9 to 46.4 in 2064/65
Total population in millions	Increases from 5.1 to 6.7 by 2064/65	Increases from 5.3 to 7.4 by 2064/65
Working-age population (15 years & above) in millions	Increases from 4.1 to 5.7 by 2064/65	Increases from 4.3 to 6.3 by 2064/65
NZS recipient age population (65 years and above) in millions	Increases from 0.8 to 1.8 by 2064/65	Increases from 0.9 to 1.9 by 2064/65
Labour force in millions	Increases from 2.9 to 3.8 by 2064/65	Increases from 3.1 to 4.3 by 2064/65
Aggregate labour force participation rate (percentage)	Reduces from 70.0 to 66.3 by 2064/65	Reduces from 71.8 to 68.3 by 2064/65

Source: Stats NZ, the Treasury

Figure 1 below illustrates the age distributions for the age groups "Under 15 years", "15 to 64 years" and "65 years and over", across 40 years in intervals of a decade, from the 2025 LTFM. Over time in the projections the eldest age group increases as a proportion of the population, while the other two age groups reduce. This is a result of increasing *longevity* and decreasing *fertility*. It also shows that the increased assumptions around numbers of migrants, of which around two-thirds are aged between 15 and 44 years in general, are not enough to offset these two drivers of an ageing population structure.

The age group proportions over the decades in this latest version of Stats NZ's population projections are actually quite similar to those of the 2020 base version that was used as the basis of demographic projections in the 2021 LTFM.

Figure 1: The age distribution of New Zealand in the 2025 LTFM projections



Source: Stats NZ, the Treasury

Economic variables and assumptions

Key Points

- *Nominal GDP* is an important economic variable in the LTFM, because it serves as a growth driver of many fiscal variables and as the denominator of key fiscal indicators.
- Little has changed in how most economic variables are projected in the LTFM compared to the 2021 version, but some stable assumptions have been revised.
- One quite different aspect is the government *10-year bond rate's* attaining its stable value in the first projected year rather than the 2021 LTFM's long transition path.

The LTFM's economic projections serve two important purposes. First, they provide information about the potential future paths of variables that are of interest to many and varied groups in society. These variables include *inflation*, the *unemployment rate* and other labour force statistics, *interest rates* and both *real GDP* and *nominal GDP*. Second, often in combination with demographic variables, economic variables are used as components in projecting fiscal variables. In particular, *nominal GDP* is not just a growth driver in modelling many fiscal variables but also provides the denominator for the majority of fiscal indicators, such as *net core Crown debt* as a percentage of GDP.

Demographic, Economic and Fiscal Assumptions and Logic in the 2021 Long-term Fiscal Model includes a section which explains in detail how the various economic variables are both projected and used in the LTFM. This Analytical Note, as it did for the demographic assumptions and variables, will focus on what has changed regarding the economic variables in the 2025 model. As Table 2 below details for several important economic variables, there actually has not been a great deal of change in the assumptions used in their projections. This is also the case for the modelling logic applied to them.

Table 2: Stable assumptions and period to stabilise in 2021 and 2025 economic projections

Economic variable	2021 LTFM Assumption	2021 LTFM Number of years to stabilise	2025 LTFM Assumption	2025 LTFM Number of years to stabilise
Annual labour productivity growth	1.0%	Attained by third projected year	0.9%	Attained by first projected year
Unemployment rate	4.25%	Attained by third projected year	4.25%	Attained by second projected year
Average weekly hours worked	33.7 hours	Attained by third projected year	33.6 hours	Attained by third projected year
Consumers Price Index annual growth	2.0%	Attained by third projected year	2.0%	Attained by first projected year
Government 10 year bond nominal annual return rate	4.3%	Does not stabilise until 20 years into the projections	4.3%	Attained by first projected year

Source: The Treasury

The *annual labour productivity growth* assumption was reduced from 1.0 percent to 0.9 percent because the most recent thirty-year average is used to set this value. When the 2021 LTFM was produced this average was 1.0 percent, to one decimal point of a percentage point. However, the average reduced to 0.9 percent when the 2020/21 to 2023/24 years replaced the 1990/91 to 1993/94 fiscal years in the thirty-year series used to calculate the 2025 LTFM assumption.

While *annual labour productivity growth* is an important input into projecting GDP, it also serves as the proxy for real wage growth in the economy. For this reason, while altering it affects tax and other revenue projections, which grow in line with GDP, the change also flows through to expenses projected under a bottom-up modelling logic.

Demographic, Economic and Fiscal Assumptions and Logic in the 2021 Long-term Fiscal Model describes *bottom-up modelling* logic in detail for the various expense and asset types to which it is applied in the baseline scenario of the LTFM. However, it may be helpful to provide a brief summary of what this terminology means. It refers to applying modelled projection logic to expense and asset types, rather than assuming their growth in projected years would come from a share of the *Operating Allowances* for expenses or *Capital Allowances* for assets. The latter approach, which is referred to as *top-down modelling*, is used in the Treasury's Fiscal Strategy Model. *Bottom-up modelling* may entail using combinations of demographic and economic growth drivers, as with health and education expenses, or involve transitioning to stable percentages of *nominal GDP*, as with *Defence expenses* and several asset types.

Expense categories, such as *Health*, *Education* and *New Zealand Superannuation*, use *annual labour productivity growth* to represent real wage growth in their projection modelling. Other expense areas, like *Defence* and *Law and order*, and asset categories like *Property, plant and equipment*, grow in line with GDP under *bottom-up modelling*, as most revenue types do in the LTFM. The effects on revenue and expenses are not symmetrical. However, they are much more aligned than in scenarios which assume that most expense categories derive their growth from capped *Operating Allowances*, such as those used in the Fiscal Strategy Model's baseline projections. This is because the growth applied to the projected *Operating Allowances* is not linked to *annual labour productivity growth*.

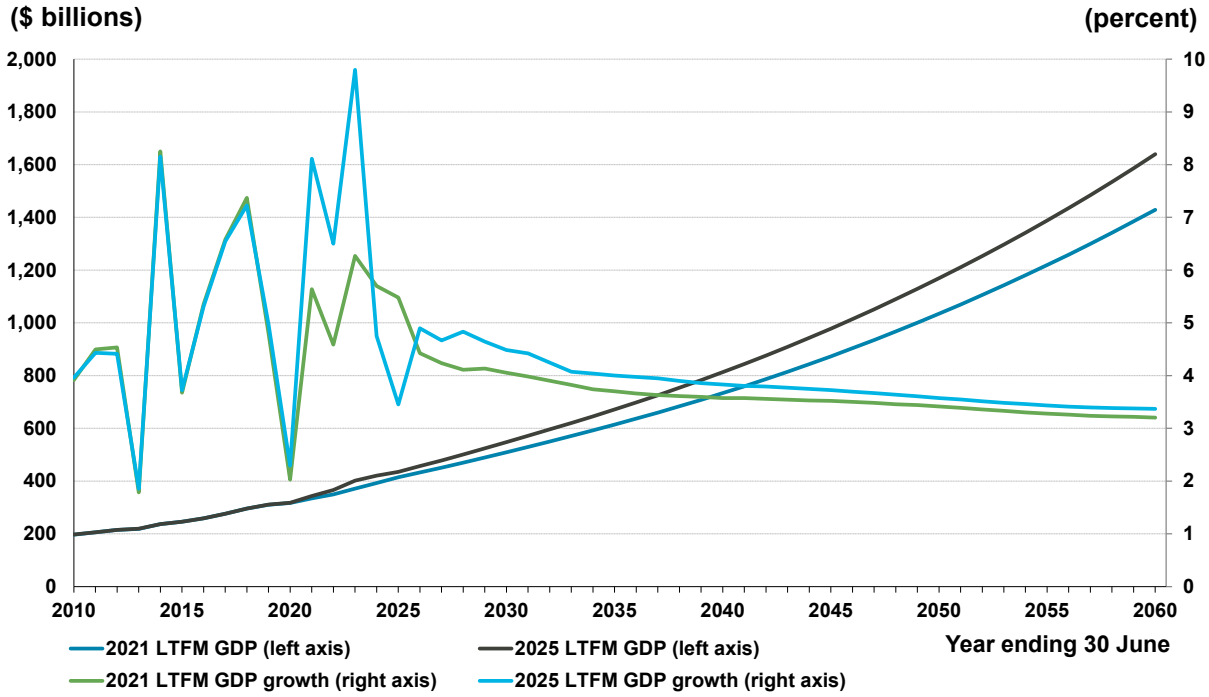
The historical average used to set the *annual labour productivity growth* assumption is affected by older years in the series being replaced and also by revisions that Stats NZ makes to both historical GDP and estimates of total hours worked in the economy. This average could rise, fall or stay the same when it is recalculated each year, around mid-September when June quarter GDP is released.

The small decrease to the assumption concerning *average hours worked* in the economy came about after discussions with the Treasury's macroeconomic forecasting team. Outturns of this variable had generally been lower than they were before the period from 2008 to 2010 that was affected by the global financial crisis (GFC). One of the reasons for this is likely to be, with an ageing population, that many older people are only working part-time. *Average hours worked* only directly influences projected GDP while it is changing so this small reduction in its assumed stable value has limited impact on the fiscal projections.

When the 2021 LTFM was produced the Stats NZ demographic and labour projections out to 2064/65 were available, which was the final year of the 2025 LTFM's projections. However, some important exogenous input tracks, like those of *Student Loans* and the Accident Compensation Corporation (henceforth just referred to by its familiar acronym, the ACC), were not produced to as far into the future as 2064/65 at that time. Consequently, the 2021 LTFM only projected its fiscal variables out to the fiscal year 2060/61.

Hence, while Figure 1 can display age distributions for both the 2021 and 2025 LTFMs out to 2064/65, later comparisons of the two models' projections of economic and especially fiscal variables cannot. The end-of-decade fiscal year 2059/60 is used in all of these situations for comparisons, which still allows a 30-year period of projections for both models between 2029/30 and 2059/60.

Figure 2: Nominal GDP levels and annual growth in the 2021 and 2025 projections



Source: The Treasury

Updating demographic data, particularly labour force projections, affects the GDP track that the LTFM produces. Figure 2 above displays the 2021 and 2025 *nominal GDP* projections aligned to the left axis, and the annual percentage growth of each GDP track aligned to its right axis.

Despite one driver of the GDP projection, namely *annual labour productivity growth*, being slightly lower in the 2025 version of the LTFM, overall annual growth of the GDP projection was a little stronger. As the *Consumers Price Index (CPI) inflation* measure is the same in both LTFM versions, this indicates that the annual growth of the *aggregate labour force*, which is the only other key driver of projected *nominal GDP*, was stronger in the 2025 version of the model than in the 2021 LTFM. However, the graph does also show the gap between the two annual growth rates of GDP narrowing as the projection proceeds, which means that the gap between the growth rates of the two labour force projections is reducing over time.

With stronger overall combined growth of its drivers and its forecast starting point in 2028/29 already above the level projected for that year in the 2021 LTFM, the GDP track produced by the 2025 LTFM rises to greater heights than that of its predecessor. While it is GDP growth that affects projections of most revenue types and many expense categories, and those of several asset and liability types too, it is the level of GDP that serves as the denominator for key fiscal indicators e.g. *net core Crown debt* as a percentage of GDP. Consequently, such a fiscal indicator needs to be considered both in terms of the levels that the variable in its numerator reaches and the effect of the bigger GDP denominator when making comparisons between its projections under the two LTFM versions.

The other main economic variable change was not to the assumption for the *government 10-year bond's* long-run stable nominal annual rate of return, which was actually unchanged, but rather to its transition path to stability. In the 2021 LTFM the end-of-forecast value of this variable was only 2.8 percent, reflecting its 1.2 percent level in 2019/20, the last outturn year at that time. While underlying economic principles suggested that this low bond rate was not the new norm for the New Zealand economy and would rise in the future, a lengthy transition path to the assumed long-run "stable" rate of 4.3 percent was built into the model. This long and slow lift to a higher annual return rate was similar to the approach being taken at that time by international agencies such as the United States Congressional Budget Office and the United Kingdom Office for Budget Responsibility.

Despite aligning with the projection methodology of respected overseas agencies, actual events soon showed that the rise in interest rates in New Zealand was not unfolding at the slow rate envisioned in the 2021 LTFM's projections. In the aftermath of the COVID-19 pandemic interest rates rose quickly, with the *10-year bond rate* already at 4.2 percent by 2022/23 and lifting further to 4.9 percent in the following year. The Budget 2025 forecasts that are the base of the 2025 LTFM's projections return it to 4.3 percent, to the nearest decimal point of a percentage point, by 2028/29, the final forecast year. Consequently, the long-run stable assumption has not been changed but as we are effectively at that value by the final year of the latest LTFM's projection's base it just remains there throughout the projection horizon.

The change in how the *bond rate* is transitioned to its stable value in the 2025 LTFM was the only significant alteration to the modelling logic used for the projection of any economic variable, relative to that which was applied in the 2021 version of the model.

Fiscal variables and assumptions

1 Fiscal variables and assumptions

Key Points

- The "exploding debt" scenario of the LTFM should not be seen as an outcome that the Treasury expects to occur, but rather as one that cannot be allowed to unfold.
- Some changes in the 2025 LTFM are due to new fiscal variables, like OBEGALx, or changes to the Crown accounts, like separating out the ETS from other provisions.
- The main assumptions behind the bottom-up modelling still result in net core Crown debt to GDP eventually rapidly rising, as it has in all of the past LTFM versions.

Projecting fiscal variables, and in particular key fiscal indicators, is the major purpose of the LTFM. These projections, unlike a forecast, may not necessarily portray beliefs about likely outcomes, but may be more designed to show a future that cannot be allowed to occur.

The *bottom-up* projection of the key fiscal indicator of financial position, *net core Crown debt excluding New Zealand Superannuation Fund (NZSF) and advances*,[3] as a percentage of GDP, is such a "warning signal" projection. Its basic premise is that New Zealand faces pressure in maintaining a sustainable public fiscal position. This is, to name just some of the more important influences, due to our ageing population structure, climate change, the requirement to replace old or weather-damaged infrastructure, and increased defence spending as a response to changing international settings. However, the response to such pressures in the projections, once they cause expenses to exceed revenue, is simply to borrow to cover the growing deficits. These rising debt levels incur ever-increasing interest costs that just accelerate the growth of the deficits and hence of debt also. A so-called "exploding debt" scenario unfolds as the projection proceeds.

However, this is not realistic. Our public finance regulations would not permit such a response to carry on year after year, it is not how our governments over the last 40 years have acted, and international lenders would either charge very high interest rates or just stop lending to New Zealand if we behaved in this manner. Furthermore, it is not posited as what the Treasury believes will occur. In other words, it is not a forecast. Rather it should be viewed as a potential pathway, which is heavily influenced by both the assumptions underpinning it and its forecast base, which must be averted. That is not to say that the fiscal pressures are not real, as they most certainly are, but rather that policy changes will be needed to ensure that the fiscal position remains sustainable. These policy changes may not all be just in the areas causing the fiscal pressures. Considerations of which groups in the population will be affected by various changes are important in deciding how to address these future challenges, and many areas of expenses, revenue and assets may be required to be altered to achieve fair and equitable outcomes.

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[3] Henceforth this fiscal variable will be referred to just as *net core Crown debt*, but it should be kept in mind that it does not include the financial assets and liabilities of the NZSF nor the core Crown advances. Other measures of the debt liability exist and will be specifically identified if mentioned.

There are over 400 rows of data and checks in the LTFM's *Fiscal Forecasts* worksheet that provides the majority of the fiscal data to the main modelling worksheets. Around 250 separate fiscal variables are input into these modelling worksheets, compared to less than twenty economic variables. The fiscal variables cover a range of different revenue, expense, asset and liability categories. As well as *net core Crown debt*, the total Crown *operating balance before gains and losses*, or *OBEGAL*, is used as an indicator of financial performance. Recently an alternative measure has been introduced called *OBEGALx* (New Zealand Government, 2024). This is *OBEGAL* excluding the revenue and expenses of the ACC, which is a Crown entity. Forecast and projected paths of both core Crown revenue and expenses are also commonly cited in publications such as the government's annual Budget.

As with demographic and economic variables, the 2021 background paper on the LTFM has detailed descriptions of how the many and varied fiscal variables are projected. Due to this the focus of this paper will again be on the changes that have occurred in fiscal projections. There are, unsurprisingly, more of those than with the economic or demographic variables.

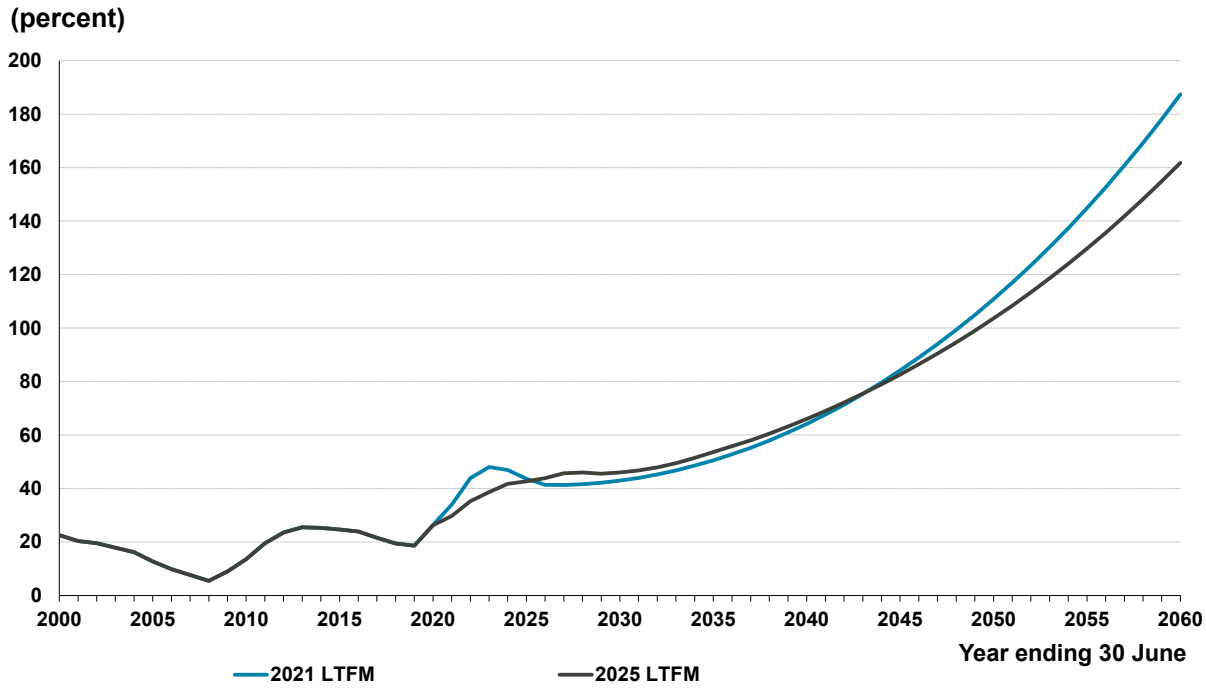
There are several reasons why the modelling of some fiscal variables has changed. There are new fiscal variables in the 2025 LTFM, such as *OBEGALx*, that were not being reported when the 2021 version of the model was produced. Accounting changes have led to revisions, such as with the *Emissions Trading Scheme* (henceforth just referred to by the acronym ETS) provision, which in the 2021 LTFM was a component of the liability type *Provisions* but is now treated as its own separate liability category.

Research and experimentation have also suggested improved methods of projecting some variables. For example, in the 2021 LTFM when *Capital Allowances* were allocated in forecast years to set the base for *bottom-up* asset projections they were all allocated to core Crown *Property, plant and equipment*. Better data on the actual distribution of these allowances, along with consultation with the Treasury's Fiscal Reporting team, has led to changes. The projection bases of core Crown *Equity accounted investments*, *non-Student Loans advances* and *Intangible assets* also now get shares of the *Capital Allowances*.

Despite new variables, altered modelling logic, changed assumptions, and an updated forecast base, one of the most significant projections of the 2025 LTFM, namely the *net core Crown debt* to GDP track, is still climbing rapidly beyond the 2040s. This is illustrated in Figure 3 below, which compares the projections of the 2021 and 2025 versions of the LTFM for *net core Crown debt* as a percentage of GDP. The 2021 track is over 25 percentage points of GDP higher by 2059/60, the final year shown on the graph. However, past the early to middle 2040s both projections are rising so quickly that only three years prior to this final year, in 2056/57, the projection of the 2021 LTFM was slightly below the percentage of GDP attained by the 2025 LTFM's projection in 2059/60.

Figure 3 helps to illustrate that, while changes to forecast bases and modelling logic affect the point at which *net core Crown debt* starts to accelerate upwards and the percentage of GDP that it reaches after 40 years, the "exploding debt" path still, under certain important assumptions, eventually unfolds. These assumptions are that, in the *bottom-up* projection, tax and other forms of revenue grow largely in line with GDP while public spending on health and pensions in particular push expenses to grow faster than this.

Figure 3: Net core Crown debt as percentage of GDP in the 2021 and 2025 projections



Source: The Treasury

2 Revenue

Key Points

- Tax is the largest revenue type and there have been changes to how some tax types are projected that have lifted tax projections compared to those of the 2021 LTFM.
- Stronger forecast bases of some revenue types, such as Sales of goods and services, have led to their projections being above those of the previous LTFM.
- Modelling of projected ETS revenue and cash receipts have been revised so that the ETS provision's decline over forecast years does not continue to an unrealistic point.

Tax revenue is the biggest form of public revenue, accounting for around 90 percent of core Crown revenue and over 70 percent of total Crown revenue. As such it is an extremely important determinant of the fiscal position and so efforts to improve its modelling in the projected years have been common over past years and Long-term Fiscal Statements.

Five major tax types were modelled in the 2021 LTFM. Each was transitioned to a stable level of GDP over the early projected years. Once this level was attained it remained at that percentage of GDP in later years and so effectively grew in line with GDP. The assumed stable levels for each tax category were based on averages of outturns back to 2006/07, with adjustments made for things like the increase in the *Goods and services tax* (henceforth referred to by the acronym GST) rate from 12.5 percent to 15 percent in 2010. The largest

tax category, *Source deductions*, which mainly consists of pay-as-you-earn (PAYE) tax on salaries, wages and transfers, had modelling logic built in that could project fiscal drag[4] over a chosen number of years, but this was not applied in the main *bottom-up* projection.

The first change that came with the 2025 LTFM modelling was the split of the tax category *Remaining tax types* into *Other personal income tax* (henceforth referred to by the acronym OPIT) and the other tax types comprising this tax category, which retained the label *Remaining tax types*. OPIT was split off because it consisted of all the personal income taxes apart from *Source deductions*, such as *Other persons tax* paid by the self-employed and small businesses and *Resident withholding tax on interest*. OPIT made up three quarters or more of the former *Remaining tax types* category and, of most significance, most of the tax types that made up OPIT were also subject to the tiered personal income tax thresholds and hence to fiscal drag. This meant that if a scenario were run where fiscal drag was to be assumed for at least some of the initial projected period then it should be applied to OPIT as well as to *Source deductions*, which would not have been possible in the 2021 LTFM. The division also allowed the greatly reduced *Remaining tax types* to be limited to an assortment of non-GST, non-transport affiliated indirect taxes, of which *Tobacco excise equivalent duty* comprised more than 40% of the total. As will be described, this decomposition allowed the modelling in projected years of the collection of taxes now comprising *Remaining tax types* to be improved.

The other changes in *Tax revenue* modelling concerned individual tax categories' projection logic, which was prompted by changes to the level of *Source deductions* relative to GDP over both outturn years since the 2021 LTFM was produced and the Budget 2025 forecast base of the 2025 LTFM. Higher inflation following the pandemic years and ongoing fiscal drag since personal tax reductions were made back in 2010, until some further tax cuts were enacted in July 2024, had both contributed to *Source deductions* rising by more than a full percentage point of GDP between 2019/20 and 2023/24. The inflation driver returned to levels closer to those of the long-run stable value of 2 percent over the Budget 2025 forecast years. However, the fiscal drag influence still operated, so that by the final forecast year of 2028/29 *Source deductions* were well above the average percentage of GDP based on data going back nearly two decades. Consequently, had it been retained, the projection modelling logic used for *Source deductions* in the 2021 LTFM would have resulted in this tax type reducing year by year relative to GDP for more than the first decade of the projections. This seemed particularly unrealistic given that, if fiscal drag was modelled,[5] *Source deductions* would rise from its forecast base as a percentage of GDP, not decline. After consultation within the Treasury, it was agreed that it was more realistic to grow *Source deductions* from its end-of-forecast level using its demographic and non-demographic drivers.

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[4] Fiscal drag occurs in a tiered tax regime where, once taxable income rises above the lowest tax threshold, the average tax rate rises with every extra dollar of taxable income. Its biggest effects occur in crossing a tax threshold, but it applies to all taxable income above the bottom tax threshold. The name is derived from its negative effects on labour supply, causing a drag on the economy, and it causes net (of tax) income to grow more slowly than gross income.

[5] Fiscal drag is not usually modelled in the projections. This is mainly because it would be unrealistic to leave fiscal drag in place over a 40-year projection. Not only would this not reflect the frequency of past changes to personal tax regimes in New Zealand, but it would also result in the marginal tax rate for someone working 40 hours per week on the minimum wage moving from a 17.5% marginal tax rate at present to a 33% rate after 40 years. The potential solution to this problem of modelling fiscal drag just for some initial number of projected years is also problematic, as it would mean "second guessing" taxation policy in terms of when fiscal drag would eventually be addressed.

Admittedly, this approach also has its drawbacks. Not modelling fiscal drag in projections effectively means that the personal tax regime is being changed in each projected year to address this effect. This goes against the LTFM's guiding principle of projecting out current policy. If that principle was strictly adhered to then the modelling would assume that the existing tax regime stays in place. However, in this case, it was judged to be appropriate, for reasons of realism and reflecting historical precedents, to override retaining current policy. The option of modelling fiscal drag for a chosen number of projected years was retained in the 2025 LTFM, but it was not used in the baseline projection.

The demographic driver of *Source deductions* is the growth of the labour force. The other driver is *average hourly growth in the nominal wage*, which in projections is modelled as the combination of *annual labour productivity growth* and *CPI inflation*. The projected *annual labour productivity growth* serves as a proxy for *average hourly growth in the real wage*.

OPIT also needs the ability to model fiscal drag if required as this tax type has similar drivers of growth to those of *Source deductions*. Consequently, it was projected in the 2025 LTFM using the same modelling logic as that applied to *Source deductions*.

Corporate tax, GST and *hypothecated transport taxes* were all still projected by transitioning them to a stable percentage of GDP and retaining them at that level in later years.
The difference for these three tax types from the 2021 LTFM modelling was that the stable percentage is now based on the average over the last five outturn years and the five-year forecast base. Using more recent years on which to base percentages, including what is predicted in the forecast years from which the projections arise, seemed more aligned to likely future paths rather than averages reflecting percentages from more than a decade ago. Overall, this revision to the projection methodology has resulted in the combined stable percentage of the three tax types reducing by half a percentage point of GDP.

Remaining tax types, now that this category no longer includes OPIT, consists of various *excise* and *excise equivalent*[6] *duties*, *customs duty* and some duties and levies on domestic goods and services. *Excise duty* on imported tobacco and both imported and domestically produced alcohol account for over 80 percent of this tax category. As both excise categories only have their rates inflation adjusted each year and tobacco consumption is declining, this tax type was projected using just the combined growth of CPI and the *working-age population*. This means that it slowly reduced relative to GDP over the projected years.

ETS revenue, which is a component of the category *Other sovereign revenue*, is now held constant. In the 2021 LTFM *ETS revenue* declined until it matched the combination of *ETS expenses* and *ETS cash receipts from auctioned units*. As the annual growth of the *ETS provision* is that year's *ETS expenses* and *ETS cash receipts* minus *ETS revenue*, the 2021 model's approach resulted in the *ETS provision* initially reducing and then staying constant at a relatively low level. The new modelling holds both *ETS revenue* and *ETS cash receipts* constant at their end-of-forecast levels while *ETS expenses* grow at the level of *CPI inflation*, resulting in the *ETS provision* still initially declining but growing from the early 2040s onwards. Holding nominal amounts constant is not an ideal option over a long projection, as their values get effectively inflated away. However, the ETS is designed

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[6] Excise duty levied on imported, rather than domestically produced goods, is called excise equivalent duty. The same tax rate is applied whether the good is imported or produced in the country.

to incentivise industries to curb their carbon emissions so *ETS revenue* may reduce over time if it is successful in achieving this objective. Hence flat *ETS revenue* and *cash receipts* tracks seem like an acceptable option in this situation.

The modelling of other variables in the *Other sovereign revenue* category has not changed much. However, the forecast base of *ACC levy revenue* is considerably higher than was envisioned in the 2021 model, due to rate increases that have occurred or been built into planning since then. This has resulted in the projected track of total Crown *Other sovereign revenue*, of which *ACC levies* are the largest component, being significantly higher in the 2025 LTFM than in its predecessor.

While the projection logic of *Sales of goods and services* at both the core Crown and total Crown levels has not altered significantly, there is a similar situation to that which occurred with *ACC levy revenue*. The forecast base of total Crown *Sales of goods and services* was well above where it was in the 2021 LTFM, resulting in a stronger projected track.

There has not been a lot of change in how components of the other three revenue categories, namely *Interest revenue*, *Dividend revenue*, and *Other revenue*, are modelled. However, especially for *Interest revenue* and to a lesser extent *Dividend revenue*, outturns and subsequent forecasts at both the core Crown and total Crown levels have been much stronger than those built into the 2021 LTFM, leading to increased projected tracks. Some of this rise will be due to the higher interest rates that have occurred since the 2021 model was produced. The *New Zealand Superannuation Fund*, which is a core Crown entity, has had returns above expectations, contributing to increases for interest revenue and dividends.

What the last three paragraphs emphasise is how strongly projections depend on their forecast base and, in terms of setting assumptions for them, recent preceding outturn years.

Table 3: Revenue categories as percentages of GDP in the 2021 and 2025 projections

Revenue category	2021 LTFM in 2024/25	2025 LTFM in 2024/25	2021 LTFM in 2059/60	2025 LTFM in 2059/60
Tax revenue	CC 27.3% TC 27.2%	CC 27.8% TC 27.6%	CC 27.6% TC 27.4%	CC 28.0% TC 27.8%
Other sovereign revenue	CC 0.8% TC 2.1%	CC 1.0% TC 2.4%	CC 0.5% TC 2.1%	CC .0.5% TC 2.6%
Sales of goods and services	CC 0.4% TC 4.7%	CC 0.5% TC 6.1%	CC 0.4% TC 4.9%	CC .0.4% TC 5.7%
Interest revenue	CC 0.2% TC 0.6%	CC 0.9% TC 1.6%	CC 0.3% TC 0.7%	CC .0.7% TC 0.9%
Dividend revenue	CC 0.4% TC 0.3%	CC 0.4% TC 0.3%	CC 0.5% TC 0.5%	CC .0.5% TC 0.3%
Other revenue	CC 0.2% TC 0.9%	CC 0.3% TC 1.0%	CC 0.2% TC 0.9%	CC .0.2% TC 0.6%
TOTAL REVENUE	CC **29.3%** TC **35.8%**	CC **30.8%** TC **39.0%**	CC **29.6%** TC **36.5%**	CC **30.2%** TC **38.0%**

Source: The Treasury

Table 3 above compares the various revenue categories, at both core Crown (denoted CC in the table) and total Crown (denoted TC in the table) level, in 2024/25 and 2059/60. The first forecast year for the 2025 LTFM is 2024/25, while it was the last one for the 2021 version. The penultimate projected year in the 2021 LTFM was 2059/60, which is close to the end of the 2025 LTFM's projection horizon.

The 2025 LTFM's core Crown and total Crown revenue projections both decline relative to GDP between 2024/25 and 2059/60. Most of the drop in the core Crown was due to *Other sovereign revenue*, where *ETS revenue* is flatlined in nominal dollars over the projection. For the total Crown both *Sales of goods and services* and *Interest revenue* reduced as percentages of revenue. Most of the decline in *Sales of goods and services* occurred across its forecast base. This was also true for *Interest revenue*, but that reduced further over the projections as its Crown entities component did not grow as fast as GDP.

By contrast the 2021 LTFM's core Crown and total Crown revenue projections both slightly increased as percentages of GDP over these years. Most categories, for both core Crown and total Crown, either lifted a little relative to GDP or stayed at a similar level. As has been described some of the 2021 LTFM's *Tax revenue* modelling was significantly altered in the 2025 version of the model. Despite these changes, the overall increases as percentages of GDP in this large revenue category were actually quite similar for the two models.

The rises in total revenue relative to GDP in the 2021 projection, compared to the reductions in those of the 2025 LTFM, mainly arise from two sources. One of these is that there have been some small adaptations to some components of non-tax revenue modelling that have resulted in some tracks, which in the 2021 model grew in line with GDP, growing a little slower than GDP in the latest LTFM. The other main reason is that 2024/25 was the year from which projections launched in the 2021 LTFM, while in the 2025 version it was followed by four more years of forecasts, in which some revenue types hardly changed in nominal dollar terms and so reduced as percentages of GDP over these forecast years.

The influence of their forecast bases on projections is further emphasised by both core Crown and total Crown revenue being at higher levels of GDP by 2059/60 in the 2025 projections than they were by the same fiscal year in the projections of the 2021 LTFM. This occurred despite the later model's revenue projections slightly declining relative to GDP between 2024/25 and 2059/60 while those of the 2021 LTFM rose. This difference in the directions of both the core Crown and total Crown revenue projection pathways between the two models was not enough to offset the impact of their projections arising from quite markedly different forecast bases. The 2025 LTFM's revenue projections had a forecast base that ended in 2028/29 at 31.0 percent of GDP for core Crown and 38.7 percent for total Crown revenue. Both of those percentages are considerably higher than the 29.3 percent and 35.8 percent of GDP for core Crown and total Crown revenue respectively in the last forecast year, 2024/25, of the 2021 LTFM.

3 Expenses

Key Points

- Just as in past LTFMs, New Zealand Superannuation and Health are the expense areas producing the greatest sources of fiscal pressure in the 2025 projections.
- Changes to the modelling of several working-age welfare types have reduced their growth rates over the projected years relative to those of the 2021 LTFM.
- Revisions to how the residual growth factors used in projections of Health and Education expenditures are estimated considerably lowered both growth drivers.

The public pension, *New Zealand Superannuation*, is included in the expense category *Transfer payments and subsidies*. *New Zealand Superannuation* is one of the two major causes, *Health expenses* being the other, of why expenses overtake revenue in *bottom-up* projections and begin the accelerating upward path of *net core Crown debt*. However, despite its key role in shaping the LTFM's projections of key fiscal indicators, nothing has altered in how *New Zealand Superannuation* is projected. Its recipient growth is still modelled by that of the "*65 years and over*" age group and its payment rates aligned to the path of *average wage growth*.

There have been changes in modelling the main working-age benefits, namely *Jobseeker Support*, *Sole Parent Support* and *Supported Living Payment*. When the 2021 LTFM was produced these benefits had their payment rates linked to average wage growth and so that was how they were modelled. Indexing of the payment rates of these benefits was returned to *CPI inflation* in 2024, which had been the policy prior to 2020. To address this change the 2025 LTFM modelled these benefits with just CPI used for payment indexation, rather than *nominal wage growth*, for the first decade of the projections. After that *wage growth* was deployed again for the remaining projected years.

While reverting to wage indexation after a decade of projections is not in line with the normal guiding logic of the LTFM, which is to "project current policy", it was done for similar reasons as to why fiscal drag was not projected for *Source deductions* and OPIT tax revenue. Retaining fiscal drag in the projections would result, after 40 years, in the unrealistic outcome of minimum wage earners being on a high marginal tax rate. Similarly, continuing to index these working-age benefits with CPI while *New Zealand Superannuation* is indexed with wage growth would produce a questionably large gap between payment rates for working-age beneficiaries and pensioners. Currently an adult on *Jobseeker Support* with no children receives about 87% of what is paid to a superannuitant getting half the *New Zealand Superannuation* couple rate. After a decade of CPI indexation in the projections, while *New Zealand Superannuation* is indexed with wage growth, this ratio reduces to 80 percent. It will stabilise at that percentage if *wage growth* is applied in the projections after that, while if indexation by just CPI is continued the ratio would fall to 63 percent by 2064/65. This is well below the 69 percent ratio reached in 2019 that was a deciding factor in moving the main working-age benefits to wage indexation.

Another change with the working-age benefits is that they were still above "normal" levels relative to population size and economic conditions when the 2021 LTFM was produced because of the effects of the COVID-19 pandemic on the labour market. This was especially true for *Jobseeker Support*. Because of this their modelling logic gradually transitioned their annual recipient numbers back to stable ratios, based on pre-COVID-19 numbers, of their relevant population groups. With a further four years having passed this was no longer necessary and recipient growth modelling was restored to just applying population growth weighted by age and gender distributions for each working-age benefit category.

There were also changes made to the modelling of *Working for Families tax credits*, *Student Allowances* and the collection of benefits and transfers labelled *Supplementary Benefits*. *Accommodation Assistance* was separated from this collection and projected as its own category using recipient growth aligned to that of the *working-age population* and payment indexation applying the "CPI changing to wage growth after a decade" logic. The remaining *Supplementary Benefits* collection and *Working for Families tax credits* were also projected with these growth drivers, while *Student Allowances* only differed in using the growth of the population aged between 18 and 40 years rather than that of the *working-age population*.

The *Supplementary Benefits* collection in the 2021 LTFM included *Accommodation Assistance*. This collection, *Working for Families tax credits* and *Student Allowances* were all transitioned to stable levels of GDP in that model, similarly to the projection logic used for a tax revenue type like *Corporate tax*. After comparing the outcomes of this projection logic with the growth rates of the benefit types concerned over recent outturn years and over the forecast base, it was considered that this modelling resulted in these benefits growing too quickly over the early projected years. As a consequence, the methods of projecting *Accommodation Assistance*, *Working for Families tax credits*, *Student Allowances* and the category *Supplementary Benefits* were modified to the new ways that have been described.

Table 4 below shows similar comparisons for welfare expense categories' projections in the 2021 and 2025 versions of the LTFM that were made in Table 3 for revenue categories. Unlike in Table 3 there is no need to distinguish between core Crown and total Crown variables because all benefits and transfers are purely core Crown.

Table 4: Welfare expense types as percentages of GDP in the 2021 and 2025 projections

Expense category	2021 LTFM in 2024/25	2025 LTFM in 2024/25	2021 LTFM in 2059/60	2025 LTFM in 2059/60
New Zealand Superannuation	5.1%	5.3%	7.6%	7.5%
Jobseeker Support	0.9%	1.1%	0.7%	0.7%
Sole Parent Support	0.5%	0.5%	0.4%	0.3%
Supported Living Payment	0.6%	0.6%	0.6%	0.5%
Working for Families tax credits	0.7%	0.7%	1.0%	0.6%
Supplementary Benefits and Accommodation Assistance	1.5%	1.5%	1.5%	1.2%
Student Allowances	0.2%	0.1%	0.2%	0.1%
TOTAL WELFARE	**9.3%**	**9.8%**	**12.0%**	**10.9%**

Source: The Ministry of Social Development, the Treasury

It is notable in Table 4 that, despite welfare expenses being half a percentage point of GDP higher in 2024/25 in the later model than in the 2021 version, the 2025 LTFM's projection has dropped to over a full percentage point of GDP lower than that of the older version by 2059/60. The two main sources of the difference between the two 2024/25 percentages of GDP, namely *New Zealand Superannuation* and *Jobseeker Support*, also contributed to the difference between the projections of the two models. The latest projection of *New Zealand Superannuation* did not increase by as much as it did in the 2021 LTFM projection, due to the growth rate of the "*65 years and over*" age group not exceeding that of the labour force by as much as it did in the 2021 projection. The 2025 *Jobseeker Support* projection reduced by more than the 2021 LTFM's projection did, mainly due to the later model using CPI indexation over the first decade.

Contributing even more to overall welfare expenditure attaining a lower percentage of GDP by 2059/60 in the 2025 LTFM projection than in that of the 2021 version were *Working for Families tax credits*, *Accommodation Assistance* and the category *Supplementary Benefits*. The modelling changes in the 2025 LTFM that have been described for these welfare expense types resulted in them growing slightly slower than GDP. For the 2021 projections the stable percentages of GDP that *Working for Families tax credits* and *Supplementary Benefits*, which *Accommodation Assistance* was part of in that LTFM, transitioned to and then stabilised at were, respectively, higher or at the same value as their percentages of GDP in 2024/25, the final year of their forecast base.

Other than *Interest expenses*, for which the modelling has not altered from how they were projected in the 2021 LTFM, the majority of non-welfare expenses have their growth over forecast years controlled by *Operating Allowances*.

In the Treasury's Budget forecasting, expense categories like *Health*, *Education*, *Law and order*, etc. do not change much in nominal dollar terms beyond the first forecast year as their combined growth over these years is expected to be limited to the item listed as *Forecast New Operating Spending* less the item listed as *Top-down Expense Adjustment*. The annual increments in the *Forecast New Operating Spending* are called the *Operating Allowances*.

In terms of assessing the impact on variables like core Crown expenses, *OBEGAL* or *net core Crown debt* there is no need to allocate the *Forecast New Operating Spending* less the *Top-down Expense Adjustment* into individual expense categories. However, to set the correct base for *bottom-up* projections allocation of this forecast growth is required. If an expense type like *Health*, which would be assumed to receive a major share of the *Operating Allowances* over the forecast years, did not include these allocations, then the projection would grow from an unrealistically low level. Furthermore, all of the growth that would be assumed to occur over the forecast years would not carry out into the projections.

With the exception of *Health* and *Education* expenses, where age and gender distributions can be applied to project recipient growth, the other spending areas are projected in a similar manner to that applied to some tax revenue types, such as *Corporate tax*. Their long-run "stable" percentages of GDP are based on the average over the preceding five outturn years and their five-year forecast base. The forecast year values of each expense type include their allocated shares of the *Forecast New Operating Spending* less the *Top-down Expense Adjustment*. This is a change from how they were estimated for the 2021 LTFM, where the values used were based on longer averages of just outturn year values. However, as with *Tax revenue* projections, using more recent percentages of GDP, including those over the forecast years, seemed more appropriate in reflecting the base of the projection.

In the 2025 LTFM *Defence* expenditure was modelled as transitioning to a 2 percent of GDP stable value. This is considerably higher than its 0.8 percent of GDP average over the forecast base and the previous five outturn years. The higher value was used because there is a stated intention to increase spending in the future by the current administration and the main opposition party has indicated that it broadly agrees with the need for higher future defence spending (New Zealand Government, 2025).

Health and *Education* expenses were still modelled as described in detail in *Demographic, Economic and Fiscal Assumptions and Logic in the 2021 Long-term Fiscal Model.* Updated demographic distribution weights from the Ministry of Health now just use five classes of *Health* spending, having dropped the *Unmet Need* category, so the updated *Health* projections incorporated that reduction in divisional categories.

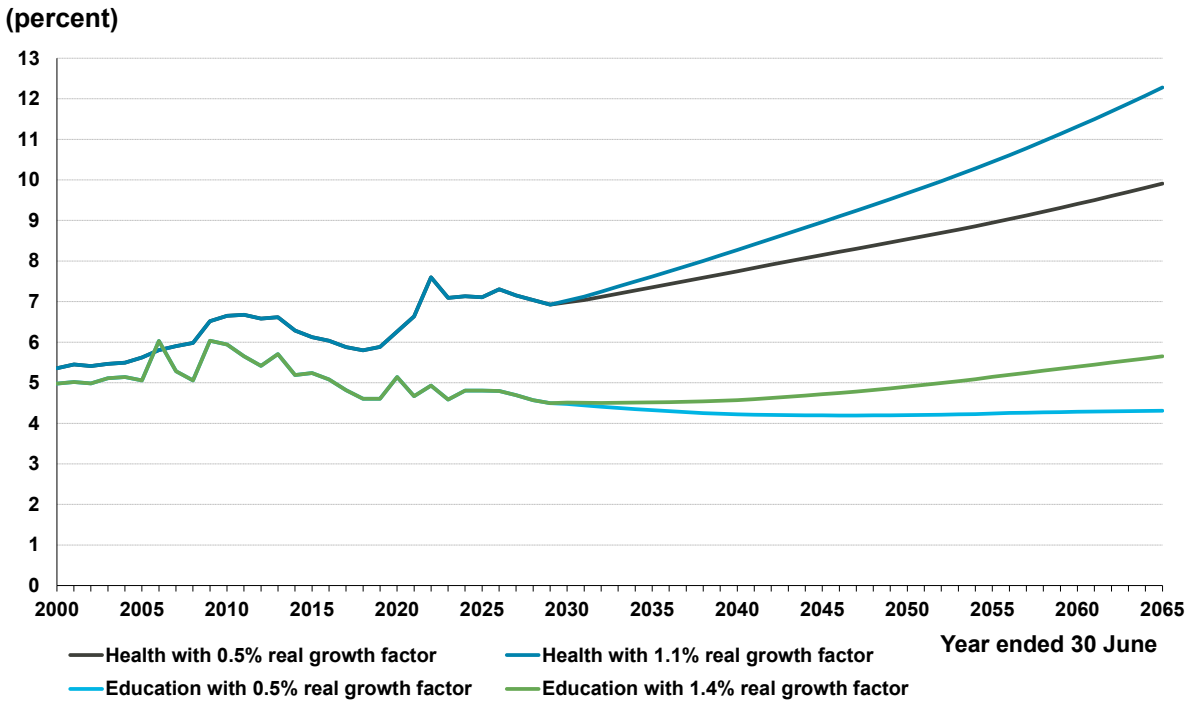
Both *Health* and *Education* expenses had their residual growth factors reassessed. These are values calculated from applying the modelling structure to outturn years and analysing what extra level of growth, applied as a constant value in each year, would minimise differences from the outturns. In the past these had been calculated as an average of the actual residual, as a percentage of the outturn's annual growth, after all other components, such as recipient growth, inflation, and labour cost changes had been accounted for. The drawback of this method was that its results were quite dependent on which outturn year was used as the base from which modelled values originated. It was decided to instead compare the historical series of outturns' growth each year to the combined growth effects of all contributing factors in the modelling equation and calculate what value for a constant residual growth factor would minimise the differences with the outturns' growth over the series. This removed the dependency on any starting point which meant that all years of outturns contributed equally to calculating the most appropriate residual growth factor.

Values of 0.5 percent were calculated as the best options for the residual growth factors of both *Health* and *Education* expenses. These estimates represented considerable reductions from the assumptions of 1.1 percent for *Health expenses* and 1.4 percent for *Education expense*s used in the 2021 LTFM.

The technique for calculating the residual growth factors was considered to be an improvement, both because the projection is based on a growth model rather than a levels-based one and that this change removed the dependency on a chosen base year. However, such big differences in the assumptions certainly warranted further checks. One way to do this was to apply the 2021 LTFM's residual growth factor assumptions to the 2025 LTFM's forecast base and compare the resulting projections, along with those produced using the updated residual growth factors, to the general direction for both *Health* and *Education* expenditure as percentages of GDP since the beginning of the century. The results are depicted in Figure 4 below.

The lower trajectory of *Health expenses* under the revised residual growth factor appears to be more in line with how health spending has actually proceeded relative to GDP over the last 24 years than the track produced using the 2021 LTFM's higher value. In 1999/00 core Crown *Health* was at 5.36 percent of GDP and had increased to 7.13 percent by 2023/24, an increase of 1.77 percentage points of GDP. If this rate of increase continued out to 2064/65 core Crown *Health* expenditure would reach 10.16 percent of GDP. While this is above the 9.91 percent attained by this year in the 2025 LTFM's projection, the difference is only 0.25 of a percentage point of GDP. Using the 2021 LTFM's real growth factor produces a 2064/65 value of 12.28 percent, which is 2.12 percentage points of GDP higher than 10.16 percent.

Figure 4: Core Crown *Health* and *Education* projection options as percentages of GDP



Source: The Treasury

Of course, the needs of an ageing population could speed up the growth of health spending relative to GDP, which could result in the higher projection ending up being the more realistic one. However, even this lower *Health* expenditure track contributes to an "exploding debt" scenario, as is depicted in Figure 3. In order to avoid rapidly rising levels of debt, while accommodating *Health* expenditure growing even faster, there would need to be changes in other areas. Public spending in at least some expense categories such as *Defence*, *Law and order*, *Core government services*, etc. would need to decline relative to the "stable" levels of GDP modelled for them, or the rise of *New Zealand Superannuation* relative to GDP would have to be curtailed via policy change, or *Tax revenue* would need to be higher as a percentage of GDP, or a combination of some or all of these things would be required.

In recent years *Education* spending has gradually reduced as a percentage of GDP. The revised residual growth factor for *Education* does result in a small decrease by 2064/65 compared to the 2023/24 outturn, namely from 4.81 percent of GDP to 4.31 percent. By contrast applying the 2021 LTFM's higher residual growth factor produces a lift to 5.65 percent. If the average annual change over the 24 years, from 4.98 percent of GDP in 1999/00 to 4.81 percent in 2023/24, were continued out to 2064/65 the result would be *Education expenses* at 4.51 percent of GDP. While this is 0.2 percentage points of GDP higher than the 2025 LTFM's projections, that is much closer than the 1.14 percentage points of GDP above this figure that applying the 2021 LTFM's residual growth factor produces by 2064/65. An ageing population structure tends to have the opposite effect on *Education expenses* to the one that it has on *Health expenses*. Over time, proportionally fewer people in the age groups that largely use education services relative to the size of the labour force could result in the demographic driver of *Education* spending being weaker than that of GDP. However, there are always counter arguments that can be made. One is that as we move into a future that is increasingly in need of skills and less dependent on "muscle", spending per student on *Education* may need to rise to ensure that our work force has the necessary skills and abilities.

Table 5 below shows the levels, as percentages of GDP, that the various core Crown and total Crown expense categories attain by 2059/60 in the 2021 and 2025 LTFM versions.

Table 5: Expense areas in 2059/60 as percentages of GDP in the 2021 LTFM and 2025 LTFM

Expense category	2021 LTFM core Crown 2059/60	2025 LTFM core Crown 2059/60	2021 LTFM total Crown 2059/60	2025 LTFM total Crown 2059/60
New Zealand Superannuation	7.6%	7.5%	7.6%	7.5%
Other transfers	4.4%	3.4%	6.4%	6.0%
Health	10.4%	9.4%	10.4%	9.3%
Education	6.3%	4.3%	6.4%	4.4%
Core government services	1.8%	1.4%	1.8%	1.4%
Law and order	1.5%	1.6%	1.5%	1.6%
Transport and communications	0.9%	1.0%	2.9%	3.2%
Economic and industrial services	0.9%	0.9%	2.6%	2.9%
Defence	0.8%	2.0%	0.8%	2.0%
Heritage, culture and recreation	0.3%	0.3%	0.8%	0.6%
Welfare administration	0.8%	1.1%	0.8%	1.1%
Primary services	0.3%	0.2%	0.6%	0.4%
Housing and community development	0.2%	0.5%	0.8%	0.8%
Environmental protection	0.3%	0.4%	0.3%	0.4%
Interest expenses	8.0%	7.2%	11.7%	8.0%
TOTAL EXPENSES	**44.4%**	**41.3%**	**55.3%**	**49.6%**

Source: The Treasury

Core Crown expenses were projected to be 3.1 percentage points of GDP lower by 2059/60 in the 2025 LTFM than for the same fiscal year in the 2021 projections. This was mainly due to a combination of reductions in the expense areas of *Health*, *Education*, *Non-New Zealand Superannuation transfers* and *Interest expenses*. The reasons behind the decreases in the first three categories have been discussed. The decreased percentage for *Interest expenses* is due to a lower debt projection in the later LTFM, and this will be covered in more detail in the section of this Analytical Note related to *Borrowings*. There are some expenditure areas that slightly offset the reductions in these areas, the most notable being *Defence* and to a lesser extent *Welfare administration* and *Housing and community development*. The reason for the higher *Defence* projection has been explained. The smaller increases in the other two areas are due to them having higher "stable" percentage of GDP assumptions because the averages used to calculate them were based on more recent outturn years and forecasts.

The 5.7 percentage points of GDP reduction in 2059/60 for total Crown expenses between the 2021 LTFM and the later version was even bigger than for the core Crown. The expense categories of *Health* and *Education* are significant contributors, as they were for the core Crown, but the biggest drop occurred in *Interest expenses*. As mentioned above, the reason for this will be explained in more detail in the *Borrowings* section. The total Crown expense areas of *Defence*, *Transport and communication*, *Economic and industrial services*, and *Welfare administration* all contributed to providing some degree of offset to this decrease.

The results displayed in Tables 3, 4 and 5 are brought together and summarised in Table 6 below, for both core Crown and total Crown measures of financial performance.

Table 6: Operating fiscal indicators as percentages of GDP in the 2021 and 2025 projections

Fiscal variable	2021 LTFM core Crown 2059/60	2025 LTFM core Crown 2059/60	2021 LTFM total Crown 2059/60	2025 LTFM total Crown 2059/60
Revenue	29.6%	30.2%	36.5%	38.0%
Expenses	44.4%	41.3%	55.3%	49.6%
Gains/(losses)	2.1%	2.0%	2.3%	2.3%
Operating balance	**−12.7%**	**−9.1%**	**−16.5%**	**−9.3%**
OBEGAL	**Not core Crown**	**Not core Crown**	**−18.8%**	**−11.6%**
OBEGALx	**Not core Crown**	**Not core Crown**	**Not calculated**	**−12.0%**

Source: The Treasury

Higher revenue and lower expenses equate to increased *Operating balances* and a higher *OBEGAL* projection, although both LTFM versions projected these variables to be negative by 2059/60. This means that for both models operating deficits were indicated in this projected future year. However, the 2025 LTFM projections for both core Crown and total Crown *Operating balances* were not as negative by 2059/60 as those of the 2021 LTFM were in this year. The 2021 LTFM's penultimate projection year was 2059/60 and this year was also close to the end of the 2025 LTFM's projection horizon.

Given that the core Crown *Operating balance* was 3.6 percentage points of GDP higher by 2059/60 in the 2025 LTFM compared to the earlier model's projection, it might be surprising that the *net core Crown debt* graph of Figure 3 did not show bigger differences between the two models' tracks. One reason that it did not were differences in how the growth of some assets were modelled, which affected their *Borrowings* projections. The next section discusses how some asset projections have changed since the 2021 LTFM was produced.

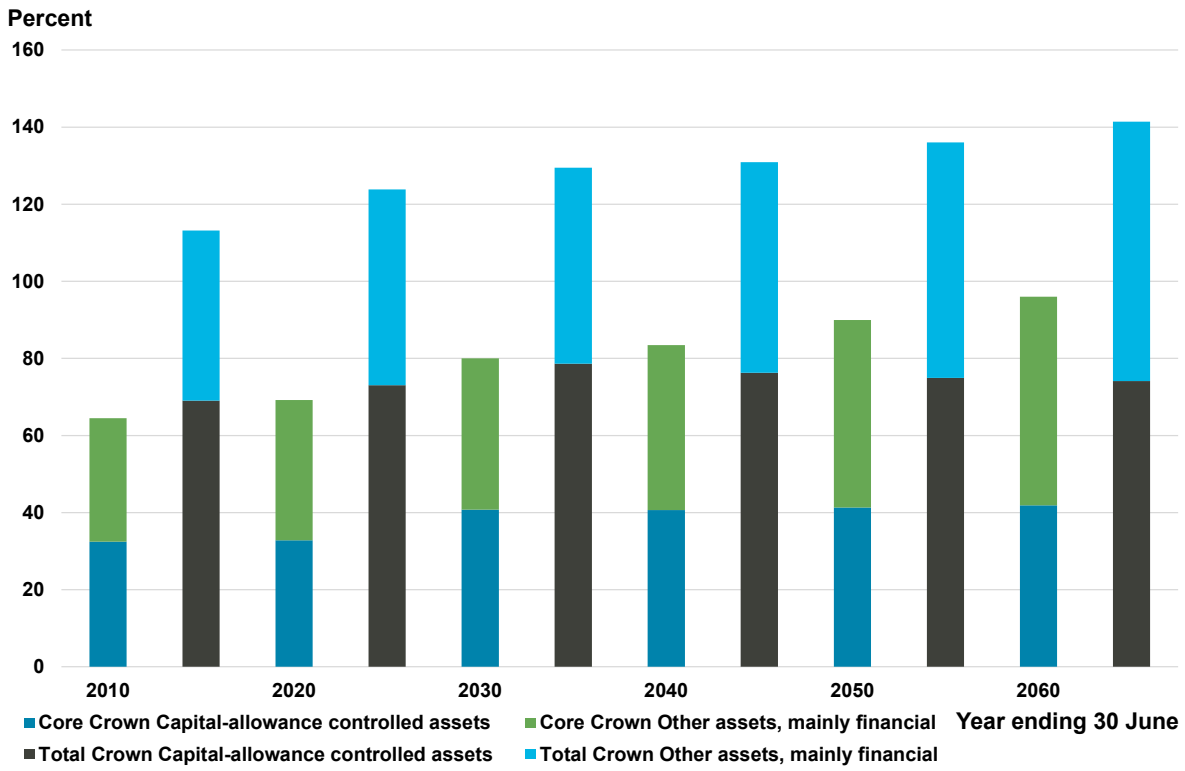
4 Assets

Key Points

- Assets are an area where research and improved knowledge of modellers has led to quite significant changes relative to the modelling applied in the 2021 LTFM.
- Core Crown asset types that are assumed to obtain their forecast year growth via shares of the Capital Allowances were the ones most affected by modelling changes.
- Projections of core Crown and total Crown financial assets were above those of the 2021 LTFM, due to higher forecast bases and stronger exogenous growth drivers.

A lot of research and experimentation has gone into improving the modelling of assets since the 2021 LTFM was produced. In recent years capital spending has increased significantly, at both core Crown and total Crown level. Furthermore, the *Capital Allowances* that are used to plan physical assets' growth over forecast years have been greater in recent Budgets.

Figure 5: Assets in two major categories as percentages of GDP in 2025 LTFM projections



Source: The Treasury

Figure 5 above depicts both core Crown and total Crown assets, divided into two major categories, over the outturn years 2009/10 and 2019/20, and for the 2025 LTFM's projected years in decade-long gaps. Capital-allowance controlled assets are those asset types for which their core Crown components, in forecasts, are assumed to derive their growth from the *Forecast New Capital Spending* less the *Top-down Capital Adjustment*. The annual increments of the *Forecast New Capital Spending* are referred to as the *Capital Allowances*.

As with the *Operating Allowances* and the expense categories which derive their growth from them over forecast years, the *Capital Allowances* need to be allocated among the appropriate core Crown asset classes in forecast years in order to set the correct base for the bottom-up projections that arise from them. The four asset classes involved are *Property, plant and equipment*, *Equity accounted investments*, *Advances excluding Student loans*, and *Intangible assets and goodwill*. The majority of the capital allowances are divided between core Crown *Property, plant and equipment* and *Equity accounted investments*, with the latter providing a sizeable share of funding for Crown entities *Property, plant and equipment*. Because core Crown *Equity accounted investments* acts as a funding channel between the core Crown and Crown entities, much of this asset's value is eliminated at the total Crown level so as to avoid double-counting of total Crown assets. *Equity accounted investments* is the only asset type where the core Crown value is larger than the total Crown value.

The Other assets grouping depicted in Figure 5 is dominated by financial assets, these being *Cash and cash equivalents*; *Receivables*; *Marketable securities, deposits and derivatives in gain*; *Student loans*; and *Investments in controlled enterprises*. The two small asset classes *Inventory* and *Other assets* have also been included in this category, although they are not financial assets. At the core Crown level the financial assets are dominated by holdings of the New Zealand Superannuation Fund, the Treasury and the Reserve Bank, while at the Crown entities level the ACC is the main holder.

Figure 5 illustrates that between 2009/10 and 2019/20 neither core Crown nor total Crown assets increased greatly relative to GDP. The 2021 LTFM, for which 2019/20 was the last outturn year, projected assets over later decades to rise at a fairly moderate growth rate. For core Crown assets the combination of the two asset categories as a percentage of GDP only got back to 2019/20 levels by 2049/50, and the total Crown value only rose slightly above the 2019/20 level by that same projected year. What actually occurred over the next few outturn years was a significant lift in both core Crown and total Crown assets relative to GDP, with the former increasing by 8.5 percentage points of GDP between 2019/20 and 2023/24 and the latter by nearly 12 percentage points of GDP over the same period. Furthermore the 2023/24 outturns for core Crown assets were more than seven percentage points of GDP higher than had been forecast in the 2021 LTFM's projection base and the increase was even bigger at the total Crown level at 16 percentage points of GDP. This increased investment since the 2021 LTFM, along with larger *Capital Allowances* being used in recent Budgets, were significant factors in reassessing the strength of asset growth, particularly for the core Crown, in the bottom-up projections.

Research and experimentation went into improving the modelling of how *Capital Allowances* are distributed among asset classes, which affected the forecast base of assets like core Crown *Equity accounted investments* and Crown entities *Property, plant and equipment*. The strengths of projected growth of several financial and non-financial asset types, at both core Crown and total Crown levels, were increased.

As with many operating expenses, there is no logical demographic distribution that can be applied to projecting asset categories, especially as they are not disaggregated to the level of items like school buildings or hospitals in the LTFM. While it is clear that the recipients of primary education services will be boys and girls of between around 5 years to 13 years of age, there is no obvious distinction between what the government spends on *Defence* in relation to, for example, females aged between 20 and 24 years and males aged

between 70 and 74 years. Likewise, there is no logical way of dividing up the expenditure on *Property, plant and equipment*, *Share investments* or virtually any assets via a demographic distribution of recipients.[7]

For this reason, many asset categories are projected in a *bottom-up* manner similar to that applied to the operating expenditure classes that do not use demographic distributions. As with those expenses, the assets are brought to stable percentages of GDP over the early years of the projections from their end-of-forecast positions and they then grow in line with the economy after that. The main difference between how this was done in the most recent model compared to the treatment in the 2021 LTFM was in how the "stable" percentages of GDP were derived. These assumed percentages were based on averages over the five-year forecast base, including the allocated shares of *Capital Allowances* for the asset classes that receive them, and the preceding five outturn years, rather than longer historical series.

Table 7: Asset categories as percentages of GDP in 2059/60 in the 2021 and 2025 projections

Asset category	2021 LTFM core Crown 2059/60	2025 LTFM core Crown 2059/60	2021 LTFM total Crown 2059/60	2025 LTFM total Crown 2059/60
Property, plant and equipment	14.0%	15.2%	54.0%	55.0%
Equity accounted investments	7.9%	17.1%	3.9%	4.0%
Advances excluding Student loans	2.8%	9.2%	10.4%	14.2%
Intangible assets and goodwill	0.5%	0.4%	1.1%	0.9%
Total Capital Allowances controlled	**25.2%**	**41.9%**	**69.4%**	**74.1%**
Cash and cash equivalents	2.7%	3.9%	4.0%	5.1%
Receivables	5.9%	6.5%	7.5%	8.0%
Mktble secs, deposits & derivs in gain	16.8%	20.1%	22.0%	26.0%
Share investments	17.2%	18.6%	22.8%	22.7%
Investments in controlled enterprises	3.8%	3.5%	3.8%	3.5%
Student loans	0.9%	1.0%	0.9%	1.0%
Inventory other assets	0.5%	0.5%	1.1%	1.0%
Total Other, mainly financial assets	**47.8%**	**54.1%**	**62.1%**	**67.3%**
TOTAL ASSETS	**73.0%**	**96.0%**	**131.5%**	**141.4%**

Source: The Treasury

[7] An exception is *Student loans*, which clearly has a recipient group heavily skewed towards people in their late teenage years through to their mid-twenties. However, the LTFM projects this asset using an exogenous track produced by the Ministry of Education which applies such demographic input.

Both the method used and the reason for its use were similar to those for the operating expenditures. As with the expenses, it was believed that an average based on a more recent set of years, including expected growth over the next few years via the forecasts, was a better indicator of likely future growth out over the longer term than an average heavily influenced by much earlier outturn years. For several asset categories their percentages of GDP in years from a decade or longer ago were quite dissimilar to more recent values.

Table 7 above provides a more detailed breakdown of the differences between the asset projections of the 2021 LTFM and those of the 2025 version. The biggest area of change is for the core Crown "Capital Allowances controlled" grouping, where this label refers to those assets whose growth over forecasts is expected to come primarily from shares of the *Forecast New Capital Spending* less the *Top-down Capital Adjustments*.

Core Crown *Equity accounted investments* and *Advances excluding Student loans* have increased significantly compared to their 2021 LTFM projections, due both to increased awareness of how they are funded via *Capital Allowances* and their marked growth over recent outturns and forecasts. The rise of this group of assets relative to GDP is not as great for the total Crown, mainly because the big increase in core Crown *Equity accounted investments* is largely eliminated at the total Crown level. This elimination is due to the core Crown *Equity accounted investments* being used to fund increased Crown entities *Property, plant and equipment*. The 2025 LTFM's modelling more accurately reflected this transfer of funding between core Crown and Crown entities assets. This linkage was missing in the 2021 LTFM, which resulted in Crown entities *Property, plant and equipment* growing quite strongly, independently of the core Crown *Equity accounted investments*.

Table 7 displays that for the majority of the asset categories, at both core Crown and total Crown level, the percentage of GDP attained by 2059/60 was higher for the 2025 projection than for that of the 2021 LTFM. For some asset types the differences were quite marked. In terms of percentage points of GDP the gap between the two models' projections for the core Crown "Other, mainly financial" assets grouping was considerably smaller than the difference for the "Capital Allowances controlled" assets grouping. However, at the total Crown level, the differences between the projections of the two models for both groupings of assets were quite similar.

For both the core Crown and the total Crown the biggest contribution to the increase for "Other, mainly financial" assets comes from *Marketable securities, deposits and derivatives in gain*, which have grown quite strongly over recent years and were forecast to continue to do so. There were also more moderate rises in most of the other financial assets.

The increased growth in financial assets is due in part to projections of some components of these assets using averages for stable percentages of GDP based on more recent outturns and forecasts. However, as well as this change to how the stable assumptions were evaluated, a number of components of various financial assets have their growth aligned to two exogenous tracks used in the models. Projected growth of some core Crown financial asset components comes from the New Zealand Superannuation Fund track, while for some components of Crown entities financial assets their projections are aligned to the ACC track. Both of these exogenous tracks were updated in the latest LTFM from the versions used in the 2021 model and both updates projected stronger growth for their asset holdings.

5 Non-debt liabilities

Key Points

- There has not been a lot of change in the latest LTFM version's modelling of non-debt liabilities relative to how they were projected in the 2021 LTFM.
- Overall core Crown non-debt liabilities reached similar percentages of GDP as they did in the 2021 LTFM, although issued currency reduced while payables increased.
- Due largely to the ACC exogenous track lifting insurance liabilities, projections of total Crown non-debt liabilities were stronger than they were in the 2021 LTFM.

Debt, or *Borrowings* as it is actually listed in the Statement of Financial Position, is a liability that will be discussed in the next section. However, while *Borrowings* are by far the largest liability that the Crown has, it is not the only form of these fiscal variables.

Little has changed regarding how the various non-debt liabilities were projected in the 2021 LTFM. One change is that the *Emissions Trading Scheme* (ETS) has become a separate liability class, when previously it was included as part of the *Provisions* liability (Ministry for the Environment, 2025). However, as a component of *Provisions* the ETS was modelled individually in the 2021 LTFM, so it has been separated from *Provisions* in Table 8 below.

Issued currency is a purely core Crown liability, so its core Crown and total Crown values are the same. *Retirement plan liabilities* are almost zero outside the core Crown, and even at the core Crown level are falling over time as the last significant public scheme, the Government Superannuation Fund, closed to new members in 1992 (GSF Schemes Administration, 2008). Hence by 2064/65 this liability, to one decimal point of a percentage point of GDP, is zero. *Insurance liabilities* are almost zero at the core Crown level and are projected as such so they are only significant at the total Crown level, where they are largely due to the Crown entity ACC and are projected using its exogenous track in the LTFM.

Table 8 – Non-debt liability categories as percentages of GDP in the 2021 and 2025 projections

Liability category	2021 LTFM core Crown 2059/60	2025 LTFM core Crown 2059/60	2021 LTFM total Crown 2059/60	2025 LTFM total Crown 2059/60
Issued currency	2.2%	1.1%	2.2%	1.1%
Payables	2.7%	3.6%	3.6%	4.8%
Deferred revenue	0.3%	0.2%	0.9%	0.7%
Emissions Trading Scheme (ETS)	0.3%	0.5%	0.3%	0.5%
Insurance liabilities	0.0%	0.0%	16.7%	23.4%
Retirement plan liabilities	0.0%	0.0%	0.0%	0.0%
Provisions	1.4%	1.2%	2.3%	1.8%
TOTAL NON-DEBT LIABILITIES	**6.9%**	**6.6%**	**26.0%**	**32.3%**

Source: The Treasury

Table 8 above shows little difference between the 2021 LTFM and 2025 LTFM projections of the core Crown combined total of non-debt liability categories as a percentage of GDP.

In the 2021 LTFM *Issued currency* stayed level as a percentage of GDP from its end-of-forecast level. However, in the latest model it was projected to grow less quickly than the economy. This reflects its reduction relative to GDP in recent outturn years and forecasts, due to less use of cash as EFTPOS and credit card transactions become more prevalent (Reserve Bank of New Zealand, 2024). The weaker projection for *Issued currency* was largely offset by a stronger track in the 2025 LTFM for *Payables*. This is due to higher outturns and forecasts of this variable in recent years, possibly again due to an increased tendency to purchase on credit.

The situation is a little different at the total Crown level, where the same swap between *Issued currency* (which is purely core Crown) and *Payables* is occurring. However, a much stronger projection of *Insurance liabilities* in the 2025 LTFM was only slightly offset by small reductions relative to those of the 2021 model in other types of liability, especially *Provisions*. *Insurance liabilities* are projected using a track provided by ACC and, as with the growth of ACC assets, their liability growth was also stronger in their updated track. This is not surprising as, over the long-term, the ACC needs to try to balance its projected liabilities, arising through accidental injuries to members of the population, with its projected assets, derived from both levies and investment income.

6 Borrowings

Key Points

- The 2025 LTFM used Cashflow Statements, at both core Crown and total Crown level to project out the changes to Borrowing, just as the 2021 LTFM did.
- Lower projected operating deficits, offset to some degree by stronger asset growth, has led to a relatively similar core Crown Borrowings track to that of the 2021 LTFM.
- Modelling changes, especially for assets projections, have helped to close the gap between core Crown and total Crown *Borrowings* seen in the 2021 LTFM's tracks.

Having now covered revenue, expenses, assets and non-debt liabilities, the next form of fiscal variable to consider is *Borrowings*. Core Crown *Borrowings* are an integral component of the key fiscal indicator of financial position, *net core Crown debt* (excluding the New Zealand Superannuation Fund and advances). That key fiscal indicator is the subject of the next section of this Analytical Note.

While the terms debt and borrowings are often used interchangeably, they do actually refer to different things. While *Borrowings* are generally the largest component of debt, and this is certainly the case in the Crown accounts of the New Zealand government, there are other fiscal variables that either add to *Borrowings* or subtract from them in forming *Gross debt*. Examples of these other interactions include the addition of *Unsettled purchases of securities* and, in the case of the government accounts, the subtraction of *Reserve Bank settlement cash and bills*. The *Debt Indicators* note in the Forecast Financial Statements provides a component-by-component difference between core Crown *Borrowings* and *Gross debt* (New Zealand Treasury, 2025c).

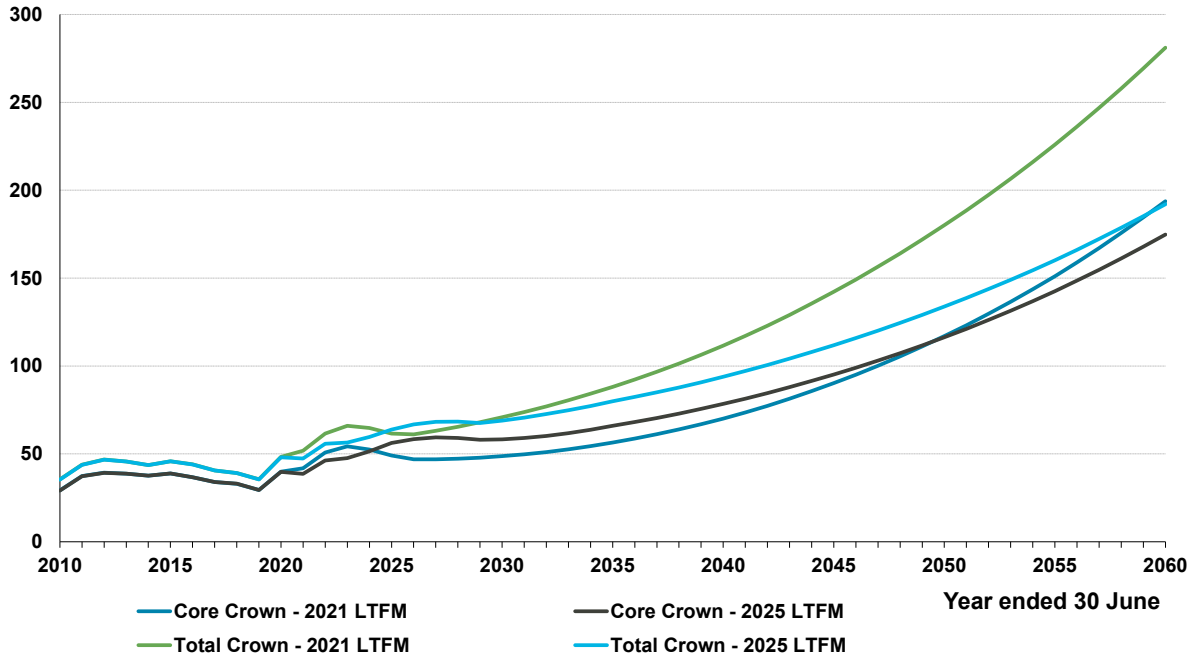
The distinction between debt and *Borrowings* is not so important in the context of the LTFM. This is because the model uses Cashflow Statements to project out the annual changes in *Borrowings*, and it is these changes which either increase or decrease the overall stock of debt in projected years.

Neither the core Crown nor the total Crown methods for modelling *Borrowings* have changed in the 2025 LTFM compared to how they were projected in the 2021 version. At both levels *Borrowings* are the residual of the modelling, meaning that all contributing variables are elements to solve the following accountancy relationship, where Δ stands for annual change:

ΔBorrowings = ΔCash – Net operating cash flows – Net capital cash flows
– ΔNet sales/(purchases) of Marketable securities, deposits and shares
– ΔCirculating currency

The 2021 and 2025 versions of the LTFM used similarly structured core Crown and total Crown Cashflow Statements to project all of the variables in this accountancy relationship in order to calculate the annual change in core Crown and total Crown *Borrowings* in each projected year. *Borrowings*, and debt, are stock variables rather than flow ones. This means that their values in a given year are not just the outcome for that year in isolation but rather are the cumulative total of the addition in that year to the previous year's total.

Figure 6: Borrowings as percentages of GDP in 2021 LTFM and 2025 LTFM projections



Source: The Treasury

Figure 6 above shows the projections of the 2021 LTFM and the 2025 version of both core Crown and total Crown *Borrowings*, as percentages of GDP. The outturn years illustrated on the graph, from 2009/10 to 2023/24, do not show a wide difference between the two tracks of *Borrowings*. However, what is quite clear about the projections of the 2021 LTFM is that the further out they proceed the more that the total Crown and core Crown *Borrowings* tracks move apart. In 2029/30 core Crown and total Crown *Borrowings*, as percentages of GDP, are just over 22 percentage points of GDP apart in the 2021 LTFM. By 2059/60 the gap between this model's two tracks had extended to more than 87 percentage points of GDP.

This widening of the gap between projections of core Crown and total Crown *Borrowings* was addressed in the 2025 LTFM because it was not reflective of their relationship in outturn years nor in their Budget 2025 forecast bases. In 2029/30 in the 2025 model total Crown *Borrowings* are just under 11 percentage points of GDP above core Crown *Borrowings*. By 2059/60 the gap has widened, but only to slightly over 17 percentage points of GDP

Modelling changes to a number of asset types helped considerably to achieving this. Keeping the projected tracks of core Crown and total Crown *Borrowings* much more aligned was not the main reason for making changes to the modelling of various asset types. As has been described in several of the explanations of asset modelling changes, these were often motivated by changes in the values of their recent outturns and forecasts relative to GDP. However, altering the projections of a number of asset types at both their core Crown and total Crown levels to better align the projected growth of core Crown and total Crown *Borrowings* was a definite consideration behind the changes. Research into the allocation of *Capital Allowances* among different asset classes not only helped to improve their projections but also assisted in bringing the two *Borrowings* tracks closer together.

The improved alignment of the projected growth of core Crown and total Crown *Borrowings* in the 2025 LTFM's projections is evident when the respective changes between the two LTFMs' core Crown and total Crown *Borrowings* tracks are examined.

In 2029/30 the 2021 LTFM's core Crown *Borrowings* track is ten percentage points of GDP lower than that of the 2025 LTFM, but by 2059/60 the earlier projection has overtaken the later one and is 19 percentage points of GDP above it. Over this 30-year period the change in core Crown *Borrowings* is 29 percentage points of GDP more than for the later model. The total Crown *Borrowings* track of the 2021 LTFM is a little higher, at two percentage points of GDP, than that of the 2025 model in 2029/30. However, by 2059/60 this positive gap has swelled to 89 percentage points of GDP. That represents a reduction of 87 percentage points of GDP in the change in the total Crown *Borrowings* track of the 2025 LTFM compared to that of the earlier model over the 30 years between 2029/30 and 2059/60. The reduction in the total Crowns *Borrowings* track between the two models, in percentage points of GDP, is three times that in the core Crown *Borrowings* projection.

These differences between the two models' trajectories of core Crown and total Crown *Borrowings* are certainly not all due to modelling changes in core Crown and total Crown assets. However, these have played an important role, particularly in reining in the growth of total Crown *Borrowings* more than that of core Crown *Borrowings* in the 2025 LTFM.

Another factor in changing the rate of growth of both core Crown and total Crown *Borrowings* and their ratio in the 2025 LTFM compared to how these were projected in the 2021 version are the updated revenue projections. As shown in Table 6, by 2059/60 both core Crown and total Crown revenue were higher, as percentages of GDP, in the 2025 LTFM compared to the 2021 LTFM. While a single year figure is not particularly significant in assessing the impacts on borrowings over thirty years the fact that revenue to GDP stays relatively level in the LTFM's projections means any differences in this year will be fairly indicative of the long-run change. The difference between the two models for total Crown revenue was about one percentage point of GDP greater than the difference for core Crown revenue. With this larger difference more or less sustained over the projected decades, changes to revenue modelling have had a greater effect on reducing total Crown *Borrowings* than on reducing core Crown *Borrowings* between the 2021 and 2025 versions of the LTFM.

Table 6 also illustrated that, by 2059/60, both core Crown and total Crown expenses were lower, as percentages of GDP, in the 2025 LTFM than they were in the 2021 model. However, unlike tax and the other revenue categories, which remain relatively stable as percentages of GDP over the projections (see Table 3), most of the major expense categories, at both the core Crown and total Crown levels, increase relative to GDP. This is true, to varying degrees, in both the 2021 and 2025 versions of the LTFM.

Table 9 below shows four major expenditure categories, plus a grouping of several smaller expense types, as percentages of GDP, for the 2021 and 2025 versions of the LTFM at both the core Crown and total Crown levels, in 2024/25 and 2059/60. The fiscal year 2024/25 was the last forecast year in the 2021 LTFM and the first forecast year in the later LTFM.

Table 9: Expense categories as percentages of GDP in the 2021 and 2025 projections

Expense category	2021 LTFM in 2024/25	2025 LTFM in 2024/25	2021 LTFM in 2059/60	2025 LTFM in 2059/60
New Zealand Superannuation*	5.1%	5.3%	7.6%	7.5%
Core Crown Health	6.3%	7.1%	10.4%	9.4%
Core Crown Education	4.7%	4.8%	6.3%	4.3%
Core Crown Interest expenses	0.7%	2.0%	8.0%	7.2%
Combined other core Crown expenses	12.2%	13.4%	12.0%	12.9%
Core Crown Expenses	**29.2%**	**32.7%**	**44.4%**	**41.3%**
Total Crown Health	6.0%	7.1%	10.4%	9.3%
Total Crown Education	5.1%	5.1%	6.4%	4.4%
Total Crown Interest expenses	0.9%	2.4%	11.7%	8.0%
Combined other total Crown expenses	18.9%	22.4%	19.3%	20.5%
Total Crown Expenses	**36.3%**	**42.3%**	**55.3%**	**49.6%**

Source: The Treasury

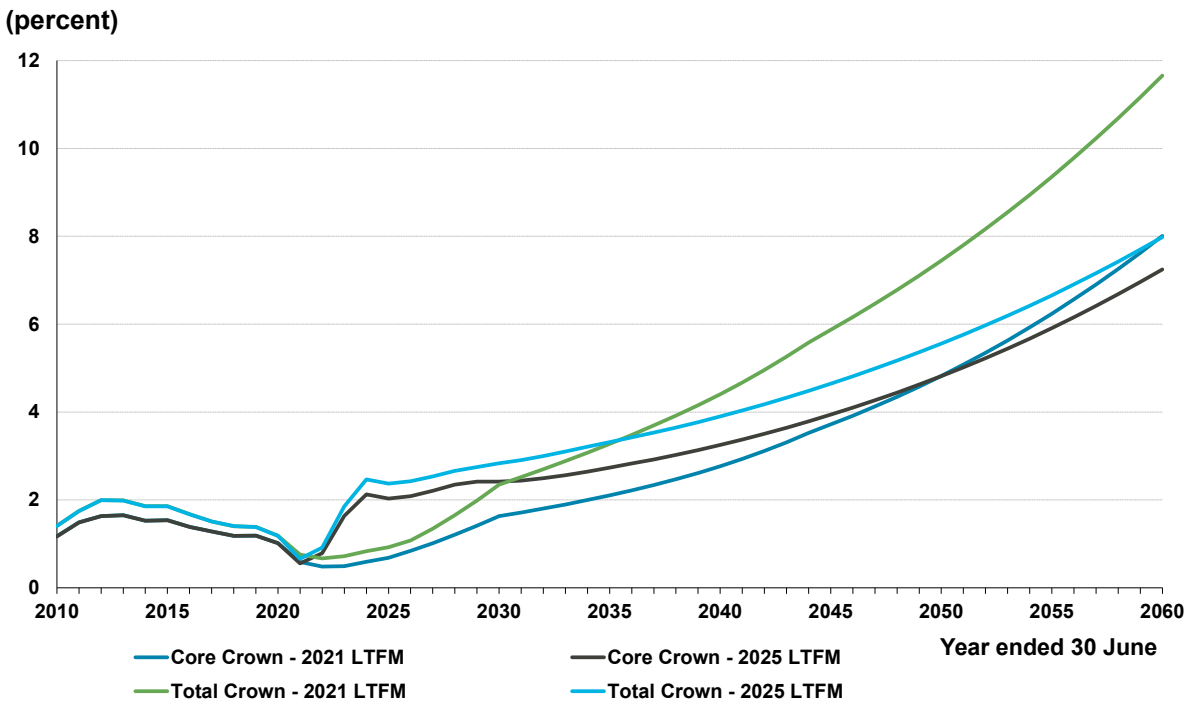
* New Zealand Superannuation is purely core Crown so is part of both the core and total Crown expenses

Much of how the expenditure projections of the two LTFM models compare, and the reasons for their differences, at both core Crown and total Crown levels, has been discussed in the Expenses section of this Analytical Note. In regard to the effects on *Borrowings* projections an important feature of Table 9 is that both core Crown and total Crown expenses were at higher percentages of GDP in 2024/25 in the 2025 LTFM but the expense projections of this model were at lower percentages of GDP than those of the 2021 LTFM by 2059/60. The big contributors to these differing paths for the core Crown expenses were *Health*, *Education* and *Interest expenses*. For total Crown expenses *Health* and *Education* expenses, along with the "Combined other" category, all contributed similarly to the reduction in the later model's projections, relative to those of the 2021 version, but nearly half the decrease was due to total Crown *Interest expenses*.

Interest expenses have not been changed in the way that they are projected as a function of debt levels and interest rates. However, as Table 9 displays, both core Crown and total Crown *Interest expenses* have increased by less, as percentages of GDP, over the 2025 LTFM's projections than they did for those of the earlier model. Furthermore, as the 2025 model assumed an interest rate on debt of 4.3 percent over its entire projections while the 2021 LTFM took the first 20 years of its projections to lift to this rate, these reductions in core Crown and total Crown *Interest expenses* are purely due to lower debt levels.

For a given interest rate as debt increases so will the *Interest expenses* that this growing stock of debt generates. These *Interest expenses* are part of the overall expenses so, in a situation like the *bottom-up* modelling of the LTFM models, where operating deficits are funded by increased borrowing, rising *Interest expenses* fuel the growth of *Borrowings* which, in turn, add to the increases in *Interest expenses*. The changes to the projections of core Crown and particularly total Crown *Borrowings* between the 2021 and 2025 versions of the LTFM reflect this relationship, except they relate to the opposite situation of *Borrowings* not increasing as quickly in the later model.

Figure 7: Interest expenses as percentages of GDP in 2021 and 2025 projections



Source: The Treasury

Figure 7 above displays the projected paths of core Crown and total Crown *Interest expenses* as percentages of GDP for the 2021 and 2025 LTFMs. The interdependency between *Borrowings* and *Interest expenses* is reflected in the similarities between the paths in Figures 6 and 7 for each of the four projections, albeit on different vertical scales for the two graphs.

As changes to the modelling of some assets, revenue categories and expense types other than *Interest expenses* reduced the rates at which core Crown and especially total Crown *Borrowings* increased in the 2025 LTFM's projections relative to those of the 2021 model, so the rates at which core Crown and total Crown *Interest expenses* grew were also decreased. Furthermore, as *Interest expenses* increased more slowly, they added to deficits at a lesser rate than they did in the 2021 projections and so helped to reduce the rate of growth of both core Crown and total Crown *Borrowings* in the 2025 LTFM.

As shown in Figure 6, the reduction in the growth of *Borrowings* in the 2025 LTFM relative to that of the earlier model was greater for the total Crown than for the core Crown. This, in turn, also reduced the growth of *Interest expenses* between the two models more for the total Crown than for the core Crown, as illustrated in Figure 7.

As depicted in Table 8, Core Crown non-debt liabilities reach similar levels of GDP by 2059/60 in both versions of the LTFM and so do not have much effect on the different rates of growth of core Crown *Borrowings*. While Table 8 does show that the later version of the LTFM lifts total Crown non-debt liabilities considerably higher as a percentage of GDP by this year, it also indicates that most of this difference occurred in *Insurance liabilities*. These have increased to higher percentages of GDP in the 2025 LTFM due to the updated ACC tracks which drive their projections, but these are mainly balancing higher ACC financial assets. Because of this they did not have a great deal of impact in dampening the growth of total Crown borrowings.

7 Net core Crown debt

Key Points

- Both additions and reductions to *Net core Crown debt* growth between 2029/30 and 2059/60 were above those of the 2021 LTFM but higher reductions slightly dominate.
- In nominal dollar terms the change in Net core Crown debt between 2029/30 and 2059/60 was quite similar for the 2021 and 2025 versions of the LTFM.
- The 2025 LTFM's higher nominal GDP projection has led to a lower Net core Crown debt track than that of the 2021 LTFM, although both rapidly rose beyond the 2040s.

Net core Crown debt is, as its name suggests, purely a core Crown fiscal variable. Consequentially, all of what has been written in this Analytical Note about total Crown revenue, expenses, assets and non-debt liabilities and how they affected the projections of total Crown *Borrowings* have no relevance to this fiscal variable.

The 2021 LTFM and 2025 LTFM projections of *Net core Crown debt* as a percentage of GDP are illustrated in Figure 3 above. This shows two tracks that are close in 2029/30, with the more recent track actually slightly higher. However, by the early 2040s this small positive gap has closed and over the next nearly two decades the 2021 LTFM's projection rises above that of the more recent model, ending in 2059/60 over 25 percentage points of GDP higher.

This is not an insignificant difference, but it is also important to note that both tracks are accelerating upwards rapidly at this point, fuelled by ever-increasing interest costs. To emphasise this, only three years before 2059/60 the 2021 track was just under the 2059/60 percentage of GDP of the later projection

The accountancy relationship that describes the variables that contribute, either positively or negatively, to the annual change (denoted ∆) in *Net core Crown debt* is, where core Crown has been abbreviated to CC and NZSF stands for New Zealand Superannuation Fund:

∆Net core Crown debt = ∆CC Borrowings + ∆CC unsettled purchases of securities
– ∆NZSF borrowings – ∆CC financial assets – ∆CC unsettled sales of securities
+ ∆NZSF financial assets + ∆CC advances

Both core Crown *unsettled purchases of securities* and *unsettled sales of securities* tend to be of similar size in forecasts and, as they are both grown with inflation over projected years, they largely cancel each other out in the growth of *Net core Crown debt*.

The core Crown financial assets removed do not include the non-cash variable *Receivables* and as *Advances* are added back this only leaves core Crown *Cash and cash equivalents*, *Marketable securities, deposits and derivatives in gain*, *Share investments* and *Investments in controlled enterprises* that are removed. As the last of these financial assets belongs entirely to the New Zealand Superannuation Fund and all of the financial assets of this entity are added back in the equation it is just the non-NZSF components of the other three core Crown financial assets listed that reduce the growth of *Net core Crown debt* relative to that of core Crown *Borrowings*.

The drivers of differences in the two LTFMs' projections of core Crown *Borrowings* were discussed in the previous section, although that did not go into providing contributions to changes in each of the tracks in nominal dollar terms. Table 10 below does provide this breakdown, as well as one in terms of percentage points of GDP, for both core Crown *Borrowings*, the major components of *Net core Crown debt*, and the other significant components of this important fiscal indicator of financial position.

With regard to Table 10 the breakdown of core Crown *Borrowings* does not correspond with how both the 2021 and 2025 LTFM projected this variable. Those models used Cashflow Statements, which are based on cash, not accrual, variables e.g. receipts rather than revenue, cash payments rather than expenses. However, the changes displayed in Table 9 do align with an alternative decomposition of *Borrowings* growth, given by the accountancy relationship:

∆Borrowings = Expenses – Revenue – Gains/(Losses) + ∆Assets – ∆Non-debt liabilities

This equation also helps to explain why revenue, valuation gains and growth in non-debt liabilities appear as negative contributions to the growth of core Crown *Borrowings* in Table 10, while expenses (both *Interest expenses* and other kinds), valuation losses and growth in assets act to increase core Crown *Borrowings*.

What is immediately obvious from Table 10 is that the vast majority of the growth in *Net core Crown debt* for both LTFM models over this 30-year period was due to the growth in core Crown *Borrowings*. Of the other components the negative contribution from core Crown financial assets growth is largely negated by the positive contribution from New Zealand

Superannuation Fund financial assets growth. This is to be expected, because the New Zealand Superannuation Fund is by far the biggest holder of *Marketable securities, deposits and derivatives in gain* and *Share investments* amongst the core Crown entities and is the only holder of *Investments in controlled enterprises*.

Table 10: Contributions to changes in Net core Crown debt between 2029/30 and 2059/60

Fiscal variable	2021 LTFM Δ in nominal $bn	2025 LTFM Δ in nominal $bn	2021 LTFM Δ in % points GDP	2025 LTFM Δ in % points GDP
Revenue	−8,162	−9,433	−469	−431
Interest expense	1,301	1,475	75	68
Other expenses	9,226	9,949	530	455
Gains/(losses)	−501	−522	−29	−24
Asset growth	710	1,136	41	52
Non-debt liabilities growth	−53	−58	−3	−3
Core Crown Borrowings growth	**2,521**	**2,547**	**145**	**117**
NZSF borrowings growth	−17	−22	−1	−1
Core Crown financial assets growth	−478	−703	−27	−23
NZSF financial assets growth	418	512	26	21
Core Crown advances growth	15	67	1	2
Net core Crown debt growth	**2,459**	**2,401**	**144**	**116**

Source: The Treasury

The other thing that stands out in Table 10 is that the total increases in core Crown *Borrowings,* in nominal dollar terms, over this 30-year horizon are quite similar for the two LTFM models. Furthermore, because the change to core Crown *Borrowings* determines the majority of the change in *Net core Crown debt*, this fiscal variable also grows by a similar amount in nominal dollar terms over both projections.

When the growths of core Crown *Borrowings* and *Net core Crown debt* are converted to percentage points of GDP the difference between the two models' projections is greater. This is because the 2025 LTFM's higher nominal GDP projection reduces all nominal dollar contributions by more than the 2021 LTFM's GDP track does over this 30-year period.

Figure 3 also shows that, from around the 2040s both LTFM's *net core Crown debt* tracks, as percentages of GDP, are rapidly rising. This, rather than any particular level of GDP attained in any given year, is the true message of the "exploding debt" projection. This cannot be allowed to occur and will need policy change to address its underlying causes before public debt reaches unsustainable levels and becomes much harder to rein in.

Conclusion

Each time that the Long-Term Fiscal Statement (LTFS) is published an accompanying Long-term Fiscal Model (LTFM) is produced to provide some of the figures, tables and graphs included in the Statement. With each update there are changes to the LTFM projections of key fiscal indicators like net core Crown debt as a percentage of GDP. There are several reasons for this, with some of the most important being an updated forecast base of the projections, policy changes, revisions to long-run assumptions, and alterations to modelling logic.

This paper has focused on how projections have changed between the 2021 and 2025 versions of the LTFM. It has discussed the demographic, economic and the fiscal variables, with the latter covered in separate sections devoted to its major categories. It has described how assumptions, accounting changes, and modelling logic were updated in the 2025 LTFM.

The real message of the projections is not what percentage of GDP that net core Crown debt reaches by a specific year. Rather it is the direction of this projection, as it accelerates upwards as operating deficits fuel debt growth, which in turn increases interest costs, which further add to the deficits and hence the growth of debt. To varying degrees, the “exploding debt” scenario has featured in all of the Long-Term Fiscal Statements of 2006, 2009, 2013, 2016, 2021 and now 2025. Sometimes net core Crown debt does not lift quite as high by the same projected year, as has happened by 2059/60 in going from the 2021 LTFM to the latest version of the model. On other occasions it is the opposite, such as in going from the 2006 LTFM to the 2009 version, where the latter model had a forecast base still affected by the impacts of the Global Financial Crisis. The 2006 projection of net core Crown debt was at 102 percent of GDP by 2049/50 but had swelled to 223 percent of GDP by the same year in the projection of the 2009 LTFM.

Whether net core Crown debt is at 100 percent of GDP, or 150 percent, or over 200 percent, is not really the point. The salient message to take is that it is well over prudent levels and that it is rapidly increasing with every further projected year. Furthermore, the underlying reason for this accelerating debt is ongoing deficits, financed by debt. Drilling down even further, the ongoing deficits are caused by expenses surpassing revenue and increasing interest costs only keep adding to these deficits. Finally, while there are various factors that put pressure on controlling expenses, such as climate change and replacing infrastructure, a dominant feature of expenses rising faster than revenue is always an ageing population structure. Two areas in particular stand out, namely New Zealand Superannuation and public health expenditure. The former is purely a product of the ageing population structure causing the recipient population of the public pension to grow faster than the labour force that helps to drive GDP. Growth in health expenses is more complicated than just being due to the ageing population, although that is a contributing factor.

Another important point to reiterate is that the LTFM projections are not forecasts. They are not what the Treasury, given the information available at the time, expects to occur. New Zealand’s history has not been one of spiralling public debt and lack of fiscal oversight, but rather the opposite, especially over the last four decades. The “exploding debt” scenario, common to all the Statements to some extent, should be seen as a warning signal. The fiscal pressures are real and need to be addressed via a range of policy change and actions. The sooner they are, the easier it will be to make changes in ways that are fair and equitable to New Zealanders of different genders, ages, ethnicities and socio-economic groups.

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Appendix: Some important equations used in the 2025 Long-Term Fiscal Model

This appendix to the Analytical Note illustrates a few of the more important equations used in the LTFM's modelling logic. It is far from an exhaustive list as there are a large number of equations used in the LTFM, particularly in the projected years, but many are simple growth calculations, summations of totals or percentage calculations.

This appendix focuses on some commonly used modelling procedures and the more complex bottom-up growth projections.

Real gross domestic product (GDP) and nominal GDP equations

$$GDP_{t+1} = GDP_t \times \frac{LF_{t+1} \times (1 - UR)_{t+1} \times AWHW_{t+1}}{LF_t \times (1 - UR)_t \times AWHW_t} \times (alpg_{t+1})$$

Where:

t and *t+1* are subscripts representing two consecutive years
GDP is real GDP in the year denoted by its subscript
LF is the size of the aggregate labour force in the year denoted by its subscript
UR is the economy-wide unemployment rate in the year denoted by its subscript
AWHW is the average weekly hours worked in the economy in the year denoted by its subscript
alpg is the economy's annual labour productivity growth in the year denoted by its subscript.

The LF multiplied by (1-UR) represents the employed labour force and multiplying this by the AWHW gives the total weekly hours worked in the economy. The division of this entire term in year t+1 divided by its value in the previous year gives the annual growth of the total weekly hours worked in the economy in year t+1. When this is multiplied by the alpg the result is the growth of the volume of hours worked combined with the growth of the output produced in those hours, which represent the overall growth of the volume of output of the economy. Put another way, it represents the growth of real (constant price, volume measure) gross domestic product.

While not strictly true in actuality, projections assume that the constant price real GDP and the current price nominal GDP differ only by inflation. Hence, nominal GDP is projected by applying annual growth equal to that of real GDP multiplied by that of CPI-based inflation for that year.

Transition to stable value equation

$$v_{t+1} = v_t + \text{MIN}(T - v_t, tr) \times \text{SIGN}(T - v_t)$$

Where:

t and *t+1* are subscripts representing two consecutive years
v is the transitioned economic or fiscal variable's value in the year denoted by its subscript
T is the targeted stable value of the variable being transitioned to this stable value
tr is the annual transition rate that is being applied, in the appropriate unit for that variable.

MIN is the Excel function that calculates the minimum value of a set of variables in the bracket.

SIGN is the Excel function that calculates the sign, -1, 0 or 1, of a set of variables in the bracket.

The MIN function ensures that any annual transition cannot be greater than the value selected in the *Assumptions* worksheet. It also guarantees that when the variable gets as close or closer to its targeted stable value than the annual transition rate would move it then in the following year it will reach the stable value. In later years the minimum value will then be zero so the variable will not change from the targeted stable value either chosen for it in the *Assumptions* worksheet or calculated from its outturn and forecast year values.

The SIGN function ensures that, whether the variable ends the forecast years above or below its targeted stable value, it will move in the right direction, be that up, down or not at all, over the transition years.

A variation on this formula for some fiscal variables, in particular the tax revenue ones that apply it, is when the variable's stable percentage of GDP, rather than its own value, is targeted. In this situation in every place in the above formula where the value of the variable in year t appears it is replaced by the value of the variable in year t divided by nominal GDP in year t. The other change is that the entire result of the formula is finally multiplied by GDP in year t+1 in order to convert the result of the transition formula from a percentage of GDP back to the value of the fiscal variable in nominal dollar terms.

Setting an average based on the final three forecast years equation

$$v_{t+1} = \text{IF}\left(y_{t+1} = \text{OFFSET}(Assumptions!\$B\$8, 0, \$C\$1), \text{AVERAGE}\left(\frac{v_{t-2}}{r_{t-2}}, \frac{v_{t-1}}{r_{t-1}}, \frac{v_t}{r_t}\right), \frac{v_t}{r_t}\right) \times r_{t+1}$$

Where:

t and *t+1* are subscripts representing two consecutive years
v is the projected fiscal variable in the year denoted by its subscript
r is the related fiscal variable, or fiscal expression, in the year denoted by its subscript
y is the fiscal year variable, denoted by its subscript.

C1 is cell C1 of the modelling worksheet which sets the number that matches the word written in cell B1 to the title of the column of assumptions in the *Assumptions* worksheet. This has $ signs either side of the column and row references of the cell so that the formula always uses cell C1 even if it is copied into a different cell or the formula is extended along a row or up or down a column.

Assumptions!B8 is the cell in the *Assumptions* worksheet that corresponds to where the first projected year, as a four digit number, is written in the column of assumptions.

IF is the Excel function that evaluates the condition at its beginning and enacts the first option following the comma if the evaluation is true or the second option if it is false.

OFFSET is the Excel function that moves from the cell nominated at its start by the number of rows following the first comma and the number of columns following the second comma.

AVERAGE is the Excel function that evaluates the average of the terms inside its bracket.

This is a relatively common technique used in the LTFM for projecting a fiscal variable from its end-of-forecast value by using a constant ratio of another related fiscal variable or fiscal expression. The constant ratio is calculated using the average of the variable of interest's ratios to the related variable or expression over the last three years of the forecast base.

What this function does is first check if the projected year, given in row six of the modelling worksheet as a four-digit number, such as 2030 to denote the fiscal year 2029/30, is the first projected year. If it is then the average of the previous three years' ratios of the variable of interest to a related fiscal variable, or possibly to a fiscal expression such as the sum or a difference of variables, is calculated.

Because this is only done if the year of calculation is the first projected one, the previous three years must be the last three years of the forecast base. If the result of this check shows that the calculation is for a later projected year than the initial one, then the ratio applied in the previous year is selected by the IF statement. For the second projected year this will be the ratio used in the first projected year, which is the average of the last three forecast years' ratios. Then for the third projected year the check will result in using the ratio from the second projected year, which is again the average of the ratios from the last three forecast years, and so on for all further projected years. In other words, the IF statement ensures that the average of the last three forecast years' ratios is applied as the multiplier in the formula in every projected year.

The equation's final part multiplies the value of the related fiscal variable, or fiscal expression, in the year of calculation by this constant average ratio over the last three forecast years to give the value of the variable of interest in that year.

The health and education expenditures' bottom-up projection equation

$$v_{t+1} = v_t \times \left(1 + \text{SUMPRODUCT}(CW, DG_{t+1}) \div \left(\frac{LE_{t+1}}{LE_t}\right)\right) \times (1 + RGF)(1 + alpg_{t+1})(1 + inf_{t+1})$$

Where:

t and *t+1* are subscripts representing two consecutive years
v is the category of health or education expenditure's value in the year denoted by its subscript
alpg is the annual labour productivity growth, a proxy for real wage growth, in the year denoted by its subscript
inf is the annual inflation, as measured by the CPI, in the year denoted by its subscript.

CW is the vector of cost weights, from the *Exogenous* worksheet of the LTFM, applied to that particular category of health or education expenditure to allot shares of demographic growth based on age and gender groups.

DG is the vector of age and gender group annual growth rates, from the *Population Treasury* worksheet of the LTFM, applied to that particular category of health or education expenditure in the year denoted by its subscript.

LE is the life expectancy, either at birth for some health categories or at age 65 for others, from the Population Treasury worksheet of the LTFM in the year denoted by its subscript (these life expectancy growth terms are only used for health, and not in the education equation).

RGF is the constant annual residual growth factor used for either all health categories or all education categories.

SUMPRODUCT is the Excel function that calculates the sum of the set of results from multiplying each pair of entries that occupy the same positions in two vectors of the same size.

This equation produces the bottom-up projection of different health and education expenditure categories. The only difference in the form of the equation is that the education categories do not have the annual growth of some form of life expectancy used as a divisor of the annual growth of the weighted age and gender groups. These divisors slow the weighted demographic growth as a proxy for healthy ageing, which is only appropriate in regard to health expenditure.

The equation is actually more complicated than it is displayed above, as there are separate SUMPRODUCT terms for female and male weighted age group growth and, in the case of health, separate life expectancy growth divisors for females and males. These parts of the equation give the growth of the recipients of these expense types, with the Cost Weights vectors based on data provided by the Ministries of Health and Education. As examples, most health categories have higher weights for older age groups while education categories are very much focused on younger people and on what stage of education they are in.

The combination of multiplying the annual labour productivity growth, a proxy for real wage growth, by inflation, as measured by the Consumers Price Index (CPI), gives nominal wage growth and hence the driver of the price component of providing these public goods.

The Residual Growth Factor is a constant term used to calibrate the backcasted values of this equation to actual annual growth for these health and education expenditure categories over the last decade or so. There are many reasons why the relatively simple equation will not accurately capture the true growth in a given year and this term is an average “catch-all” to attempt to account for these many and varied differences from what the equation’s other terms produce.